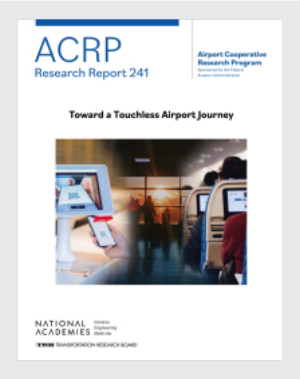
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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 241

Toward a Touchless Airport Journey

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TRB TRANSPORTATION RESEARCH BOARD

2022

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP RESEARCH REPORT 241

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FOREWORD

By Theresia H. Schatz
Staff Officer
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ACRP Research Report 241: Toward a Touchless Airport Journey is a quick reference guide for all types and sizes of airports, including their third-party vendors, and identifies readily available touchless technology features with near-term solutions and best practices to allow a safe and efficient journey. The Guide presents solutions from emerging and existing technologies to nontechnological solutions. For each solution, the Guide describes the current level of readiness, barriers to expect in implementation, and specific tips on getting started. It also looks at technology gaps and the need for medium- and long-term solutions with further development opportunities that could minimize touchpoints. This Guide suggests mitigation strategies and solutions for processes that cannot be touchless, such as wheelchair service providers and baggage tags.

The research included current best practices of airports, airlines, associations and governing bodies, service providers, concessionaires, parking and ground transportation providers, etc., and cost-effective solutions, including nonaeronautical revenue opportunities, passenger-centric smartphone and sensor technologies and applications along with their associated barriers to adoption. The Guide includes signage and wayfinding opportunities to facilitate the touchless experience and the use of technology to reduce dwell times through shorter queue lines. It looks at the customer experience and different forms of communication and the accommodation of passengers with disabilities, including those with an intellectual or developmental disability. The Guide also includes techniques for continuous disinfectant processes and global practices and innovations within and outside of the aviation sector that have been effective in making processes more efficient and touchless.

As technology continues to evolve, airports are seeking opportunities to create a touchless experience along the passenger journey. The COVID-19 pandemic has accelerated the need to eliminate many common touchpoints, which are potential points of disease transmission. This research examined the airport processes and functions that could potentially benefit from touchless technologies.

ACRP Project 11-02/Task 41 was led by Barich, Inc., in association with Airbiz Aviation Strategies, Ltd.; Decision Services International, LLC; Airport Process Design, Ltd.; and Entro Communications, Inc. The research was completed quickly to address ways an airport can be a touchless or frictionless and seamless journey experience.

Accompanying the Guide is a solution worksheet called "Taking Action," which is organized by varying points in an airport journey that an airport can use to help decide which solution to consider and implement. The worksheet is available at the National Academies Press website (www.nap.edu) by searching for *ACRP Research Report 241: Toward a Touchless Airport Journey*.

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CHAPTER 1

Setting the Stage

Introduction and Purpose

The world seems a lot smaller these days. Especially within the United States, air travel is so readily available and relatively affordable that, over the last few years, airports and air carriers have experienced higher passenger numbers than they have ever experienced. At the same time, technology adoption and customer service capabilities have grown to such a degree that even the average consumer expects a top-notch, seamless experience throughout their buyer journey—or in this case, their airport journey. As things continue to progress, any step that can potentially be automated or eliminated is up for consideration.

Enter the COVID-19 pandemic; things changed dramatically within the aviation industry. Because the virus spread so easily in groups of people, air travel drastically decreased in a very short span of time. Airports, air carriers, regulatory agencies, and solution providers began scrambling and working together to identify solutions that could protect their customers—passengers and employees alike—and prove that air travel was still a very safe mode of transportation. Any step that required customers to touch a surface, interact with another person, or spend extended periods in congested areas was now under consideration and heavy scrutiny.

But is this all that new? Anyone in the aviation industry for the last 20 years has heard of or even participated in efforts to provide a "seamless" or "frictionless" experience along the passenger journey. These efforts prioritized minimizing touched surfaces, eliminating or streamlining interactions with staff members, and reducing crowds in queuing areas. What changed is the addition of a business driver (customer health) and a higher level of urgency (reacting to an ongoing pandemic and record-low passenger numbers). Most entities interviewed for this quick reference guide stated that, for the most part, the pandemic only accelerated the need for solutions toward which they were already moving. Seamless and frictionless now include "contactless" and "touchless," and customer safety initiatives now include customer health measures.

This quick reference guide identifies readily available touchless solutions that airports and their stakeholders can implement to move further toward a touchless airport journey. The guide presents these solutions along the areas of the airport journey, and it includes a range of solutions from emerging and existing technologies to nontechnological solutions. For each solution, the guide describes the current level of readiness, barriers to expect in implementation, and specific tips on how to get started at your airport. The guide also presents a series of overall best practices and lessons learned compiled from airports that have already made progress. The goal is to offer tangible tactics for the aviation industry today, knowing the industry will continue to fill the gaps in readiness and overcome the barriers to move closer to a fully touchless airport journey.

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Research Summary

This guide is based on the efforts of a research team who reviewed literature and conducted case study interviews with large airports and airlines, medium to small airports, airport stakeholders, and relevant solutions providers. The literature research encompassed existing aviation research guidance, industry reports and white papers, news and web articles, and presentations. However, as a quick reference guide, a comprehensive U.S. airport survey that would provide a fully exhaustive look at solutions across the industry is not included.

The case study interview effort touched the following entities:

- Airports Council International – North America Business Information Technology (ACI-NA BIT) Touchless Working Group
- Alaska Airlines (AS)
- Carnival Cruise Line
- Delta Air Lines (DL)
- Digital Aviation Research and Technology Center (DARTeC) at Cranfield University
- dormakaba
- EASIER
- Embross
- Flagship Facility Services
- Gatekeeper Systems Inc.
- Gerald R. Ford International Airport (GRR)
- Golden Triangle Regional Airport (GTR)
- HMSHost
- Hong Kong International Airport (HKG)
- ICM Airport Technics
- Idemia
- Immune Gloves
- International Air Transport Association (IATA) and the IATA Passenger Accessibility Process Task Force (PAPTF)
- Johnson Controls
- Louisville Muhammad Ali International Airport (SDF)
- Lyft
- Materna
- Minneapolis-St. Paul International Airport (MSP)
- NEC
- Newark Liberty International Airport (EWR)
- Park Assist
- Pittsburgh International Airport (PIT)
- Raleigh-Durham International Airport (RDU)
- Seattle-Tacoma International Airport (SEA)
- Spirit Airlines (NK)
- Transportation Security Administration (TSA)
- United Airlines (UA)
- U.S. Customs and Border Protection (CBP)
- Vision-Box
- Winnipeg Richardson International Airport (Canada) (YWG)

In addition to these entries, the literature review uncovered information about touchless technology practices from solution-provider Bags Inc. and airports (listed alphabetically by IATA airport code) including the following:

- Hartsfield-Jackson Atlanta International (ATL)
- Abu Dhabi International Airport (AUH)
- Boston Logan International Airport (BOS)
- Bristol International Airport (United Kingdom) (BRS)
- Baltimore/Washington Thurgood Marshall International Airport (BWI)
- Charles de Gaulle Airport (CDG)
- Charlotte Douglas International Airport (CLT)
- Cincinnati/Northern Kentucky International Airport (CVG)
- Dallas Love Field Airport (DAL)
- Ronald Reagan Washington National Airport (DCA)
- Denver International Airport (DEN)
- Dallas Fort Worth International Airport (DFW)
- Detroit Metropolitan Airport (DTW)
- Dusseldorf Airport (Germany) (DUS)
- Key West International Airport (EYW)
- Fort Lauderdale-Hollywood International Airport (FLL)
- Haneda Airport (Japan) (HND)
- William P. Hobby Airport (HOU)

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- Dulles International Airport (IAD)
- George Bush Intercontinental Airport (IAH)
- Indianapolis International Airport (IND)
- John F. Kennedy International Airport (JFK)
- Kansas City International Airport (KCI)
- Harry Reid International Airport (formerly McCarran International Airport) (LAS)
- Los Angeles International Airport (LAX)
- LaGuardia Airport (LGA)
- Gatwick Airport (United Kingdom) (LGW)
- Heathrow Airport (United Kingdom) (LHR)
- Lyon Airport (France) (LYS)
- Manchester Airport (United Kingdom) (MAN)
- Orlando International Airport (MCO)
- Chicago Midway International Airport (MDW)
- Miami International Airport (MIA)
- Milwaukee Mitchell International Airport (MKE)
- Melbourne Orlando International Airport (MLB)
- Myrtle Beach International Airport (MYR)
- Oakland International Airport (OAK)
- O'Hare International Airport (ORD)
- Narita International Airport (Japan) (NRT)
- Philadelphia International Airport (PHL)
- Phoenix Sky Harbor International Airport (PHX)
- Reno-Tahoe International Airport (RNO)
- San Diego International Airport (SAN)
- San Antonio International Airport (SAT)
- San Francisco International Airport (SFO)
- Singapore Changi Airport (SIN)
- Norman Y. Mineta San Jose International Airport (SJC)
- Salt Lake City International Airport (SLC)
- Sacramento International Airport (SMF)
- St. Louis Lambert International Airport (STL)
- Stuttgart Airport (Germany) (STR)
- Tampa International Airport (TPA)
- Tucson International Airport (TUS)
- Edmonton International Airport (Canada) (YEG)
- Montreal-Trudeau International Airport (Canada) (YUL)
- Winnipeg Richardson International Airport (Canada) (YWG)
- Toronto Pearson International Airport (Canada) (YYZ)
- Zurich Airport (Switzerland) (ZRH)

For airports reviewed in the literature research, the guide only references the specific solutions noted in respective pieces of literature and, therefore, may not provide a full listing of all solutions at each of these airports.

State of the Industry

The colossal effect of the COVID-19 pandemic on passenger numbers and revenue cannot be overstated; however, most organizations interviewed noted that not much has changed within their overall strategy relative to touchless solutions. That is not to say they made no changes in response to the pandemic—all airports implemented measures such as social distancing stickers, signage, and health-related announcements over public address systems. It is to say that organizations have realized that the trends across what they were already pursuing aligned very well with the need for touchless solutions.

Three overall trends, displayed in Figure 1, rise to the surface when reviewing individual initiatives on touchless solutions by airports across the world.



Figure 1. Trends in aviation industry initiatives.

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4 Toward a Touchless Airport Journey

The following pages dive into each of these areas, describing these trends, along with COVID-19 changes, collective industry initiatives, and industry directives. However, for a full description of touchless solutions, see Chapter 2: "Touchless Solutions Across the Airport Journey Areas."



Trend Area 1: Providing a Seamless or Frictionless Airport Journey

This trend includes airport implementations of solutions such as advance reservation systems, enhanced check-in and bag-drop kiosks, facial recognition at check-in and bag drop, security, and immigration control.

What COVID-19 Changed

For most of these solutions, the change generated by the COVID-19 pandemic was more of one to repackage existing solutions as "touchless." For example, parking reservation systems are not just important because they ensure you have a spot: they are also important because many enable you to scan a quick response (QR) code to open parking gates instead of touching a button that is infrequently cleaned. They may also eliminate the need for staff interaction when leaving the garage.

Another change was that airports had increased urgency for accelerating initiatives. For example, touch-free kiosk solutions have been discussed for some time; however, new initiatives to move all touch-related actions to the passenger's mobile phone were accelerated by the vendor community. These solutions are now deployed in several U.S. airports.

Collective Industry Initiatives

The aviation industry does very well coming together to solve the challenges that individual airports are facing and then to develop best practices and guidance that the entire industry can use. The industry has come together in several ways to identify how to develop an integrated and frictionless approach across the entire passenger journey—from home to destination.

End-to-End Passenger Journey Initiatives: One example is the International Air Transport Association's (IATA) One ID initiative (https://www.iata.org/en/programs/passenger/one-id/). This uses a single identification to eliminate repetitive processes across the passenger journey, relying on a growing suite of technologies that are interoperable and scalable. Another example is New Experience Travel Technologies (NEXTT), a joint initiative between the Airports Council International (ACI) and the IATA focused on off-airport activities, advanced processing technology, and the use of interactive decision-making. This includes increased use of biometrics and mobile solutions. The goal is "to ensure the transport of passengers, baggage & cargo benefits from the latest technology developments to improve customer experience, reliability and efficiency" (*1*).

Reacting Specifically to the COVID-19 Pandemic: The focus is to leverage existing initiatives with new technologies to share travel health requirements (e.g., quarantine, vaccination, testing) using a health pass (i.e., Travel Pass, CommonPass, and smart vaccination certificates).

Industry Directives

There are several relevant industry directives (e.g., guidelines or regulations) that relate to the nature of the touchpoint and to what extent it can be completed without a human interface.

Entry/Exit Initiatives: U.S. Customs and Border Protection (CBP) was instructed to biometrically confirm visitors in and out of the United States. As technologies evolved, facial comparison—now known as the Traveler Verification Service (TVS)—has become the preferred solution. As a result of the pandemic, the CBP expanded the vision to adopt an end-to-end,

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seamless passenger journey, including self-bag-drop and security checkpoints. Canada Border Services Agency (CBSA) also introduced a primary inspection kiosk (PIK) using facial matching with an ePassport and fingerprint for pre-enrolled travel visa holders. Biometrics at immigration (and emigration) have been deployed for over a decade in some countries with the process becoming increasingly touchless with the preferred use of facial recognition technology.

Reacting Specifically to the COVID-19 Pandemic: The focus was to promote existing touchless facilities in guidance from the Centers for Disease Control and Prevention (CDC) and other airport health guidelines.

Trend Area 2: Enabling Passengers to Take Control of Their Journey



This trend encompasses solutions such as pre-airport check-in, prepayment for checked baggage, self-bag tagging, self-bag-drop, self-ordering kiosks for dining and concessions, and food and beverage mobile ordering and delivery. The use of mobile devices has increased the number of passengers wanting to make reservations online or use their mobile app to reduce the printing of boarding passes or other documents. Airports and airlines have improved their websites and apps to provide information that passengers need to make their decisions pre-arrival or by using self-service functionality. This trend also includes concepts related to passenger data sharing, specifically that passengers should be in control of the data they would like to share in return for more seamless service options.

What COVID-19 Changed

Airports repackaged several of these solutions as "touchless." For example, because self-service and mobile solutions can reduce the number of interactions between people, airports that had those solutions quickly produced marketing messaging and signage to show their customers that they had these solutions available. Many who were already considering these solutions but were not yet able to implement them sought to move more quickly.

New solutions have come about, such as virtual queuing and lights to indicate congestion of areas such as holdrooms. Both give passengers more decision-making power. Though these solutions existed in other industries, specific uses in aviation have come to the forefront since early 2020.

Collective Industry Initiatives

The collective industry focus is to identify solutions that provide passengers with greater control over their journey without privacy and safety trade-offs as well as more options to manage exceptions and remove barriers.

Digital Identity: Digital travel credentials allow an individual to assert their identity, online or in person, to the required level at every process step while maintaining the privacy of personal data.

Accessibility Initiatives: In an airport environment, assisting passengers with disabilities usually implies close physical contact between the assistance agent and the passenger. Autonomous wheelchairs and other accessible touchless features such as wayfinding apps, pet relief areas access, and touchless/intelligent lift call buttons are examples of measures aimed at removing barriers.

Reacting Specifically to the COVID-19 Pandemic:

- On accessibility, the IATA Passenger Accessibility Process Task Force (PAPTF) drafted the IATA Guidance on Accessible Air Travel in Response to COVID-19 to give coordinated guidance to airline members.
- ACI-NA established several working groups to focus on specific topics of interest in response to the pandemic. As such, two working groups that directly relate to the touchless journey

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6 Toward a Touchless Airport Journey

include the Virtual Queue Working Group and the Touchless Working Group. The Virtual Queue Working Group was formed to determine a set of business cases and associated requirements that airports could use as a guide if they wanted to implement a virtual queue. The Touchless Working Group is tasked with collaborating to discuss potential touchless solutions and the associated business cases.

Industry Directives

An enhanced passenger journey is often associated with an exchange of information. This information can be delivered upstream or off-airport and is protected by data privacy regulations.

Digital Identity: Privacy regulations, such as General Data Protection Regulation and data protection provisions, are key considerations and constraints in the implementation of identity management platforms.

Mobile Processing: The CBP released the CBP One mobile app in 2020, which allows travelers the ability to apply for an I-94 (arrival/departure record), view an existing I-94, and check wait times from their mobile device. The new app eliminates the need to fill out a paper I-94 form. CBSA introduced a mobile app to prefill customs questions ahead of accessing PIK kiosks.

Reacting Specifically to the COVID-19 Pandemic: CBSA expanded the use of its mobile app, ArriveCAN, to enable passengers to share their quarantine plans, test results, and vaccine certificates.



Trend Area 3: Providing Cleaner and More Modern Airport Facilities

This trend encompasses solutions such as touchless restroom fixtures, robotic floor cleaners, improved air filtration and ventilation, monolithic floor and wall surfaces that clearly show when they are clean, wider restroom entrance hallways, and wider open common areas that provide more natural light.

What COVID-19 Changed

These solutions naturally support a touchless approach, which led airports to repackage them as "touchless" and put effort into making sure their customers knew they existed. For some airports, urgency for existing initiatives increased, such as autonomous cleaning robots and ultraviolet (UV-C) light for disinfection. Not only did the pandemic force airports to make progress, but it also eased the introduction of these sorts of solutions to an industry that may have been hesitant about the possibility of bad press for "replacing cleaning staff with technology."

New solutions in this trend area include the following:

- More frequent and visible cleaning procedures (whereas, before the pandemic, the goal was to clean when the buildings were less occupied).
- Monitoring solutions that count the number of people using restrooms to indicate when the restrooms will likely need to be cleaned.
- Alternative elevator approaches (e.g., wave-to-call buttons).
- Alternatives to touching screens, such as interaction with kiosks through the mobile phone.
- Automatic escalator handrail cleaning solutions or antimicrobial coatings for fixed handrails.

Collective Industry Initiatives

The collective industry focus in this area has been more of one in reaction to the COVID-19 pandemic in the following examples:

- ACI's Airport Health Accreditation program is based on guidance from the International Civil Aviation Organization's (ICAO) Council Aviation Recovery Task Force (CART) Take-off

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guidance and ACI's report *Aviation Operations during COVID-19 – Business Restart and Recovery*, which are consistent with Collaborative Arrangement for the Prevention and Management of Public Health Events in Civil Aviation (CAPSCA) recommendations for states.
- The Global Biorisk Advisory Council STAR facility accreditation from the International Sanitary Supply Association is based on a demonstration of outbreak prevention, response, and recovery.

Industry Directives

Directives on safety and health requirements are at the core of aviation. The pandemic has created a rapidly changing environment where directives are constantly updated to reflect the latest research and data.

Reacting Specifically to the COVID-19 Pandemic: ICAO's CART, ACI's Airport Industry Recovery Advisory Panel, and the U.S. Department of Transportation (Runway to Recovery) led the development of best practices, recommended guidelines, and associated audit programs. The use of existing and emerging touchless technology is one of the pillars of the proposed guidelines.

All three of these trends have carried into and through the COVID-19 pandemic. It will be important to observe which of these newer or modified solutions will stand the test of time and simply become part of how the industry operates.

What It Means to Go Touchless

Moving toward a touchless airport journey encompasses reducing the three touchpoint types illustrated in Figure 2.



Figure 2. Touchpoint types.

Any solution that can (a) eliminate the need for a touchpoint or (b) mitigate the duration or extent of a touchpoint is considered a touchless solution. For example, SEA has implemented a virtual queue solution called the SEA Spot Saver program. This program allows passengers to prebook an appointment in advance on the website or at the airport using a QR code. The appointment allows the passenger to have a specific time to go through security, which helps reduce the time spent in congested security queues.

Best Practices and Lessons Learned

When implementing touchless solutions, airport operators should consider the following best practices and lessons learned:

Connect with Airport Strategy or Technology Direction
Implementing touchless solutions should be closely linked to the airport's overall strategy on passenger facilitation and technology implementation.

Understand the Importance of Strong Leadership
The complexities of implementing change combined with the financial challenges currently facing the industry require strong leadership by airports. Leadership should outline a clear corporate strategy that will be supported by internal and external stakeholders alike. In the context of COVID-19, adaptability and flexibility will be the keys to ensure the seamless implementation of new and evolving travel and health requirements.

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Collaborate with Your Stakeholders
The airport journey involves various stakeholders, each looking at implementing touchless solutions. Collaboration is essential to promote the use of consistent and holistic touchless solutions rather than a succession of standalone products.

Focus on Helping the Passenger
Most airports interviewed highlighted the importance of looking at touchless solutions from the perspectives of passengers or staff members rather than through a technical lens. The use of personas by airports emphasizes the need to help different types of passengers along their journeys. Helping travelers with disabilities is especially important.

Communicate with the Passenger
The success of touchless initiatives is largely dependent on passenger (and staff) buy-in. Measures that require a significant shift in passenger behaviors should be introduced with a clear communication plan to promote adoption. While COVID-19 provided a significant argument for change, this opportunity may lapse as we exit the pandemic and health measures are progressively phased-out.

Leverage Additional Resources in the Community and Beyond
Since the airport is an extension of the community, a key to the success of touchless solutions is to ensure that they are also consistent with practices beyond the airport. For example, resources from advocacy groups should be used to ensure that passengers with disabilities, reduced mobility, or service animals can experience a similar or enhanced level of service as that experienced by passengers without disabilities, reduced mobility, or service animals.

Understand the Importance of Data
Qualitative and quantitative data provide important feedback on the issues to be resolved and the efficiency of touchless solutions. The use of data is also key to managing demand and maintaining appropriate occupancy levels at each touchpoint by adjusting resourcing in real time where possible.

For more detail, see Appendix A "Helping the Passenger," which provides mitigation strategies for travelers who need additional assistance. See Appendix B "Communicate with the Passenger," which discusses signage best practices.

What Factors to Consider

Airport operators should consider the following factors when implementing touchless solutions:

Cost
The cost of the solution will be key in terms of the capital costs for the solution itself and the enabling works that might be required to support it or operating costs to maintain this solution. On the other hand, benefits such as reduced operating costs, delayed capital costs, or additional revenues should be tallied to provide a full picture of the effects of touchless solutions on the airport's bottom line.

Support and Maintenance Requirements
Throughout the implementation, the airport should address the ongoing maintenance of the touchless product or service and the associated need for support through internal resourcing or external contracting.

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Alternative Solutions
The implementation of touchless solutions should focus on the effect on the passenger journey rather than on the solution itself. As such, alternative solutions using different enabling technologies and with different costs, benefits, and constraints should be assessed.



Readiness
Readiness for implementation should be assessed against the airport's existing layout and systems to identify any gaps. Solutions should also consider regulatory barriers that may prevent the full leverage of some technologies based on local or international regulations.

The Value of Touchless Solutions

The assessment of touchless solutions should consider various factors including the following qualitative and quantitative value propositions:



Health and Safety
Because of the COVID-19 pandemic, the priority of taking measures to ensure the health of passengers and employees is more front and center than before. Touchless solutions should demonstrate that they enhance the health and safety of passengers.



Customer Experience
Touchless solutions should maintain, but ideally enhance, the passenger experience at the airport.



Revenue Generation
As mentioned, when assessing cost-benefit, the potential for revenue generation should be investigated as part of the implementation of touchless solutions. While this may be driven by new touchless commercial opportunities, this could also be because of increased dwell time in commercial areas due to reduced wait time induced by touchless solutions in other areas such as check-in and the security checkpoint.



Operational Efficiency
The investments in touchless solutions may also yield operational benefits that should be assessed and quantified where possible to support a cost-benefit assessment when comparing options.



Customer Accessibility
Touchless solutions should be equitable and accessible to all. Where adaptation is required, touchless solutions should demonstrate that they maintain or improve access for all passengers, rather than create additional segregation.



Space Optimization
Many touchless solutions are aimed at optimizing spatial requirements by eliminating queues and touchpoints. The extent of space savings or delayed expansion requirements should be quantified to support a thorough cost-benefit assessment.

Continue to Chapter 2 to view solutions that align with these value propositions.

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CHAPTER 2

Touchless Solutions Across the Airport Journey Areas

Airport Journey Areas

Operators of airports of all sizes have found strong value in walking through the passenger and employee journey across their airport while seeking ways to improve efficiency or the customers' experience. Several operators interviewed for this guide went through this exercise specifically to identify touchpoints and opportunities to provide a more touchless journey. Some went a step further, identifying "personas" (such as business travelers versus leisure travelers) using their overall customer demographics and mapping several distinct journey profiles for each.

Airport journey areas, identified in Figure 3, consist of several touchpoints for passengers, employees, and other airport patrons.

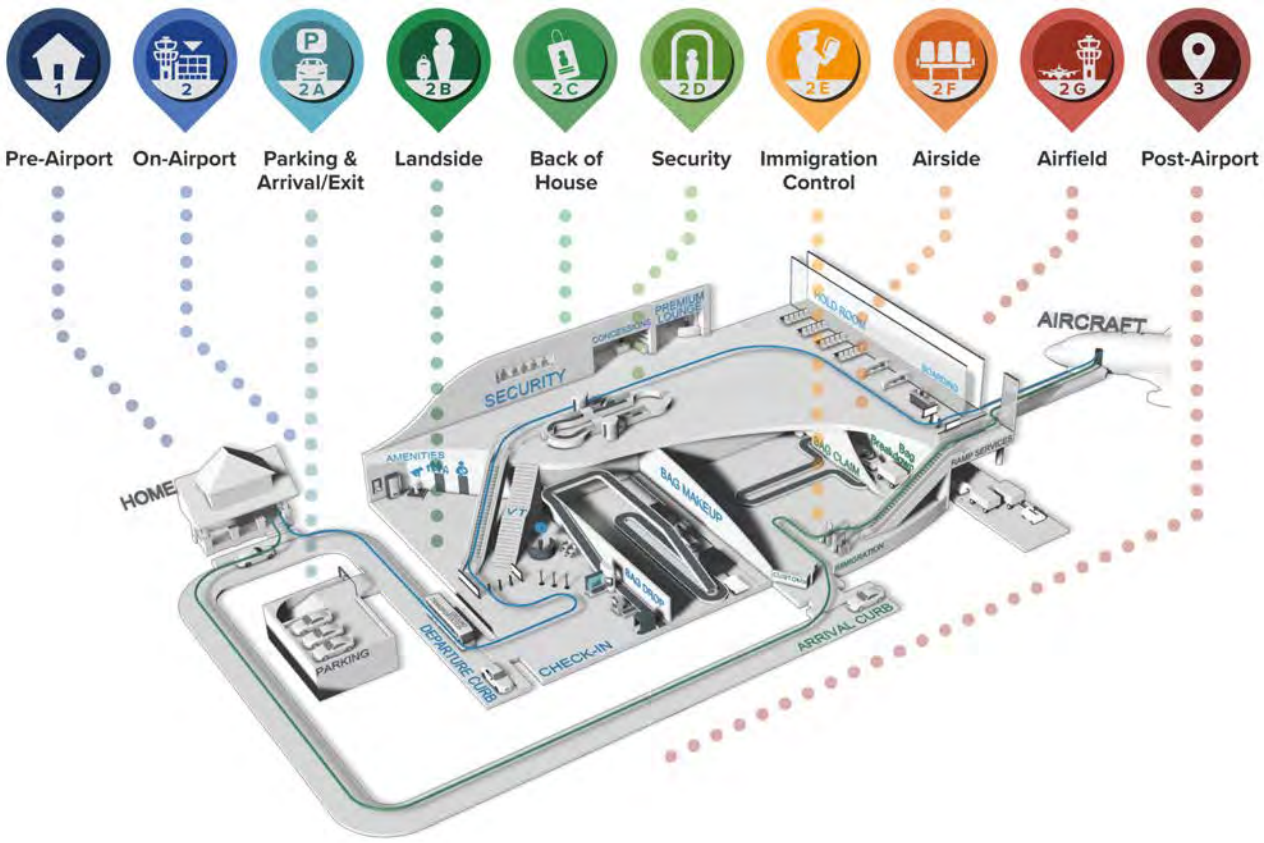


Figure 3. Airport journey areas.

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Summary of Solutions Along the Airport Journey Areas

There are many ways to break down the airport journey. Figure 4 provides the illustrations that represent all areas of the airport journey that customers may encounter, accommodating for both passengers and employees. Number 1 indicates the Pre-Airport, number 2 indicates On-Airport areas, and number 3 is for Post-Airport. Table 1 then presents a summary of solutions that can apply to each area.



Figure 4. Illustrations representing the airport journey areas.

Table 1. Solutions along the airport journey areas (ordered alphabetically).

Solution Name	Journey Areas
Access Control Systems	
Automated Access Systems	
Automated Car Park Systems	
Autonomous Wheelchairs	
Baggage Divesting Solutions	
Baggage Prepping Solutions	
Baggage Screening Systems	
Biometric Technology	
Elevator-Related Solutions	
Enhanced Surface-Cleaning-Related Solutions	
Feedback Devices	
Food Ordering and Delivery	
License Plate Recognition	
Online Service Booking	
Parking Entry and Exit Kiosks	
Parking Pay-on-Foot Kiosks	
Parking Reservation Systems	
Passenger-Assistance-Related Solutions	
Personal Hygiene and Health-Related Solutions	
Queue Management Systems	
Remote Bag Drop	
Restroom-Related Solutions	
Retail Ordering and Delivery	
Self-Service Bag-Drop	
Self-Service Kiosks	
Signage	
Smart Building Systems	
Travel Reservation (TNC, Taxi, GT, Rail)	
Travel Health-Related Solutions	

(continued on next page)

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Table 1. (Continued).

Solution Name	Journey Areas
TSA Credential Authentication Technology	
TSA Enhanced Advanced Imaging Technology	
Virtual/Digital Queue Solutions	
Water Bottle Filling Stations	

NOTE: TNC = transportation network companies (e.g., Lyft and Uber); GT = ground transportation; TSA = Transportation Security Administration.

Walkthrough of the Airport Journey Areas

Expanding on Table 1, the following provides a walkthrough of each airport journey area and identifies which touchless solutions apply to each. This section also aligns solutions with the value areas (Figure 5) discussed in Chapter 1. Each section uses the icons in shown in Figure 5 to identify the value areas addressed by the solution.



Figure 5. Value areas of touchless solutions.

Note that some technology solutions may appear multiple times in different journey areas (e.g., self-service kiosks) because although they are the same basic technology, they have varying functions in different portions of the journey. Also, note that Implementation Highlights came from interviews with stakeholders unless indicated otherwise through endnotes.





Pre-Airport

Solutions that specifically apply to when a passenger is at home or work, in a car, at the train station, or other areas leading to their arrival to the airport property.

The pre-airport stage is a critical area contributing to a touchless experience that is only growing as technology and service models mature and passengers grow more accustomed to pre-airport activities. Solutions enable passengers to either remotely complete airport processes

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or take steps ahead of traveling to reduce or eliminate touchpoints once at the airport. Many solutions involve providing information about services and products available at the airport that can improve guest experience and generate revenue.

Parking Reservation Systems (Pre-Airport Registration)

Parking reservation systems enable passengers to prebook and pay for parking before they arrive, eliminating the need for a payment interaction with a cashier or kiosk at the airport. Additional functionality allows the passenger to select the type of parking (e.g., standard, premier, electric), receive prebooking discounts, and reserve additional services such as car washing and detailing.

Technology Readiness: Existing and mature technology.

Barriers: It is necessary to educate the public that the technology is available to reserve their parking spot before arriving at the airport.

Implementation Highlights: MSP implemented this system to lower the number of parking kiosk touches and agent interactions. **PHX** began implementing the parking reservation system before COVID-19 to simplify operations in the parking facility; now, all facilities are included in the reservation system. Airports such as **RDU**, **PIT**, **SFO**, and **SEA** have also implemented these systems.

Getting Started: For airports wanting to get started but needing to consider a tight budget, the first suggested step is to have a conversation with your parking management firm. Laying out a long-term strategy can help justify costs and a strategy for purchasing a base system with room to grow. At a minimum, users should be able to reserve a spot in the parking facility and receive a QR code to use when entering and exiting the parking facility. If your airport has an installed parking and revenue control system, the infrastructure costs are minimal. Most of the costs are system and configuration services.

Virtual/Digital Queue Solutions (Pre-Airport Registration)

Virtual queue (VQ) solutions can be used to turn any physical queue into a digital queue. To date, VQ solutions are most notably applicable to queues for security; however, they can be useful at all queuing locations throughout the airport facility, such as check-in or immigration. Solutions that allow for entry into the VQ before airport arrival can reduce the amount of time a passenger must stand in a physical line, which eases congestion at the checkpoint.

Technology Readiness: Existing and emerging technology.

Barriers: It is necessary to educate the public that technology is available to reserve their time to go through security before arriving at the airport. VQ technology is not new; however, its usage in airports is new. You should expect "bumps in the road" as you begin to deploy.

Implementation Highlights: SEA (*2*) and **LAX** (*3*) have recently trialed virtual queuing solutions that offer online reservations for their security checkpoints, and SEA has funded theirs as a concept through 2022. **DEN** (*4*) piloted VeriFLY, which allowed health-conscious passengers to prebook time online for a separate screening lane. Most recently, CLEAR allows passengers to reserve a time slot online for security at **MCO** (*5*). Although it is currently on hold due to the pandemic, **YUL** was the first airport to offer online reservations for security in 2014.

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Getting Started: Start by opening discussions with airline stakeholders. Pre-airport entry into a VQ will most likely begin with an airline. So far, only a handful of airlines have participated in airport pilots, including **AS** and **DL**. Of those that have, all have come back to say this is the way of the future. Understanding your key airline directives will help you get started more quickly. Initial VQ systems impact the user's mobile device and the airport or airline websites. Since the systems will be driven by applications that link the user to information (APIs), most of the costs are system and configuration services. However, early deployments require airport staff to manage entry to the physical queue; automation of this process will require technology solutions such as eGates and biometrics, which increase airport infrastructure and system costs. Additionally, a VQ solution improves its efficiency as queuing data are integrated; if your airport does not have a queue management system, this too may be an additional cost item.

See the "Security" section of the journey for more information on VQ technology.

Baggage Divesting Solutions

Remote Bag Drop	Baggage Pickup or Delivery

Some solutions allow passengers to check baggage before airport arrival. This includes using a baggage delivery service or dropping off bags at remote bag-drop locations, allowing passengers to streamline the check-in process or bypass the check-in lobby altogether.

Technology Readiness: Existing technology.

Barriers: While baggage delivery services are convenient, most require pick up as much as 24 hours in advance for a fee.

Implementation Highlights: Passengers going to **HKG** can check their bags using the In-Town Check-In Service at the Mass Transit Railway. Bags, Inc. allows passengers to check luggage directly from the pier of their cruise ship, meaning they can travel to the airport without the burden of baggage.

Getting Started: Baggage divestment before airport arrival is an option that can be tailored based on passenger needs. For example, a remote bag drop at a nearby airport hotel frequently used by passengers may be a good location to focus on initially. Remote bag drop has also been used in locations such as cruise ports, resorts, and convention centers.

Biometric Technology (Pre-Enrollment)

Airports and jurisdictional agencies have been considering and trialing several use cases for biometric technologies to remove some of the physical and manual airport processes. Biometrics can be obtained from a passenger's face, eyes, or finger; however, the facial biometric is most often used in the United States. Most of these technologies involve some level of pre-airport enrollment from the passenger. The technology behind the facial biometric enrollment captures an image of the passenger's face, creates what is called a "digital identity" that is not an actual image of the person, then destroys the initial image taken—all in a process that can take only a handful of seconds.

This digital identity can be used throughout the passenger's journey to automate several manual processes to improve the customer experience and remove touchpoints. For example, by scanning the facial biometric, a check-in kiosk could retrieve a passenger's reservation, a coffee shop could access the passenger's payment method, and the airline could automatically board

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a passenger without scanning a boarding pass. All of this supports a touchless experience, and it would start at the pre-airport stage of passenger enrollment into the program.

However, it is important to note the following barriers to a fully seamless biometric airport journey experience.

- First, since there is not yet a common industry-wide standard for creating and storing the digital identity, solution providers in this space have created proprietary methods that do not automatically align with each other. On the jurisdictional front, the CBP has established its TVS as its means for biometric identity. Although the Transportation Security Administration (TSA) and the CBP have discussed potential shared uses, no decisions have been made. For other airport and airline implementations, solution providers each have their own means of producing the biometric identity. This creates the potential challenge of passengers needing to enroll in several similar programs to experience the full set of services offered at an airport.
- Second, there are regulatory barriers related to privacy concerns at the federal level and in many states. Some regulations were developed with conventional processes in mind that do not necessarily align with biometric processes.

Though the aviation industry has several challenges to overcome, this guide presents potential solutions that airports that are willing to lead can consider. The guide also mentions several pilot programs that are currently seeking to identify solutions to common challenges.



Technology Readiness: Existing and emerging technology.



Barriers: Detailed above.



Implementation Highlights: ATL, **DL**, and **TSA** (*6*) are currently running a facial biometrics trial that allows an end-to-end digital identity experience for a touchless experience at bag drop, security, and boarding. Passengers must have TSA PreCheck, be a Delta SkyMiles member, check in online, and provide passport information before heading to the airport. Other Implementation Highlights for later areas of the journey are noted in corresponding sections of this guide.



Getting Started: An ePassport provides a means to use biometric processes without additional pre-airport steps. Regardless of where an airport may have a priority consideration for using biometrics (e.g., self-bag-drop, security, or boarding), it is recommended to have a conversation with prospective solution providers and key stakeholders to discuss the long-term potential of an end-to-end journey as well as how the means of capturing the biometric identity will hinder or promote such a journey process.



Online Service Booking



Club/Lounge	Meeting Space	Traveler with Disabilities Registration	Wheelchair	Business Center	Car Detailing

Online booking of these services can streamline the passenger's experience at the airport. Some services apply more to the airline, but some could also be provided by the airport. For example, those who need wheelchair service and assistance from the entry to the aircraft can benefit by booking this on the airline's website. Airport operators can offer lanyards for travelers with disabilities or who need additional assistance through online reservations and communications through emails and phone calls to the airport. Recently, some airport operators have made food and beverage, retail, and duty-free purchases available online or via a mobile app to allow passengers to pick up items on the way to the gate or have items delivered to their gate upon arrival.

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Technology Readiness: Existing and mature technology.

Barriers: Educating the public that these technologies are available.

Implementation Highlights: MLB has implemented sunflower lanyards that recognize travelers who may have cognitive issues. **PHX** and **SEA** have implemented a similar service. Airports such as **MCO** have a program that allows travelers who need assistance to experience traveling in advance of their actual travel plans. Related to food and beverage, **HMSHost** (*7*) is working with partners to enable passengers to order food before they get to the airport.

Getting Started: Airports can initially target specific services and include reservation alternatives through their website for passengers and/or their airport apps. Airports need to review their existing websites to create links for easy access to reservation alternatives for these services.

Passenger-Assistance-Related Solutions

Website Chatbots	Mobile Apps	Social Media

These tools are great ways to share information on the touchless options an airport operator can offer to its customers. For example, recently, airport operators and airlines have used social media to broadcast new programs such as cleaning protocols, travel requirements, and real-time construction and traffic information

Technology Readiness: Existing and mature technology.

Barriers: There are no general barriers to these solutions. However, regarding messaging, it is important to say only what needs to be said to ensure that the message is communicated to a diverse audience properly. One potential regulatory barrier relates to freedom of speech through social media if the airport monitors comments and removes offensive or inappropriate remarks.

Implementation Highlights: MSP and several other airports have used social media to communicate changes in the terminal (including touchless options) and respond to customer questions.

Getting Started: Airports should initially set up a system of monitoring websites, apps, and social media to determine whether the information is being heard by the traveler and if the message is clearly communicated. Airports have developed specialists who review social media comments, respond, and create social media campaigns for specific issues. Most airports have most of the technology in place. However, the ongoing responsibility for monitoring specific communication technology sites must be acknowledged, and the division held responsible for managing the comments and ensuring the communication is effective.

Travel Health-Related Solutions

The IATA (*8*) offers the Travel Pass app to allow passengers to create a digital passport that provides a digital version of verified COVID-19 test results and vaccine information. It allows passengers to get information on where they can get tested and vaccinated and securely sends

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the results to the airlines and border authorities, reducing the need to stand in long queues to have paper documents verified by an agent. CBP's Mobile Passport Control (MPC) (*9*) app allows passengers to submit passport and customs declaration forms, which eliminates the need for a paper form or the use of an Automated Passport Control (APC) kiosk.

Technology Readiness: Existing and emerging technology.

Barriers: Rapidly evolving domestic and international regulatory requirements may impact solution implementation.

Implementation Highlights: The IATA Travel Pass is being trialed by over 50 airlines outside of the United States. MPC is in use at 30 airports with recent pilots at **IAD**, **PHL**, and **FLL** (*10*) this year.

Getting Started: Since this topic is rapidly evolving, airport operators should monitor U.S. and international initiatives aimed at streamlining travel and health checks. CBP's MPC is an app that was launched in 2014 and is available nationwide at international airports.

Honorable Mention Solutions

The following solutions may not be as valuable as those already mentioned, or they may already be ubiquitous across the industry.

- **Home-printed bag tags and electronic bag tags** are great opportunities for passengers to simplify the onsite check-in process. **AS** offers online self-tagging where the passengers can print baggage tags from home. British Airways offers a reusable digital bag tag that allows passengers to skip the lines at check-in.
- **Baggage prepayment** has become an expanded function of online check-in for most airlines, allowing passengers to pay for checked baggage at the same time they check in online.
- **Baggage wrapping services** may be used to reduce the number of touches by others.
- Transportation network companies (TNCs), taxi, ground transportation, and rail services allow passengers to **prebook reservations from home**. While the airports may not directly implement these solutions, they may facilitate an on-airport interface by identifying pickup/drop-off points, providing geofenced staging areas, or offering other tools to enhance the passenger experience.

These solutions bring up an important best practice for airports to consider that apply to any touchless initiative: collaborate with your stakeholders! Being 100 percent in step with the options stakeholders are offering enables you to communicate these options to your customers.





On-Airport

Solutions that generally apply to all common areas located on the airport campus.

This section includes solutions that can generally be found in multiple areas of the airport journey (e.g., elevators and restrooms). Instead of repeatedly identifying them in each area, they are described once here.

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Elevator-Related Solutions

Homing	Automatic Scheduling	Remote Calling	Touchless Buttons	Foot Pedal Buttons

Since elevators require users to interact with physical buttons, some airports have implemented solutions that reduce or eliminate required touchpoints. One way to do this is to alter the elevator's operating patterns through homing or automatic scheduling. Homing entails setting some elevators to always return to a certain floor with the doors open so passengers do not have to use the elevator call buttons. Automatic scheduling is programming an elevator to continuously run from floor to floor on an automatic schedule. The other way to reduce touchpoints is through alternative buttons such as remote calling, touchless buttons, or foot pedals. Remote calling features use Bluetooth or a QR code to enable a passenger to view the elevator control interface on their mobile phone. For QR code solutions, the code is placed above the elevator buttons, and when scanned, it sends users to a website on which they can interact with the elevator. Touchless buttons contain motion sensors to recognize when a user waves their hand in front of the button (Figure 6).

Technology Readiness: Existing and emerging technology.

Barriers: Some solutions may not be intuitive. Others require a personal device, which not all users have.

Implementation Highlights: PIT (*11*) installed wave-to-call features for outside their elevators with options for "wave for up" and "wave for down." Inside the elevators were options to wave for each floor; however, they did not remove the option to touch the standard elevator buttons. **YWG** is considering the remote elevator feature calling not only for the touchless benefits but also to aid travelers with disabilities. **MSP** implemented the elevator homing approach. **LHR** has installed elevator button covers. These covers are self-cleaning and use an antibacterial formula.

Getting Started: Airports can initially implement elevator homing or automatic scheduling as a low-tech alternative. For any option, it is important to clearly communicate these changes through signage placed on doors, walls, and stanchions in elevator waiting areas.





Figure 6. PIT wave-to-call touchless elevator buttons (photos by Pittsburgh International Airport).

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Restroom-Related Solutions

Touchless Fixtures	Increased Circulation Areas	One-Way Airflow	Restroom Use Monitoring

Restroom remodeling has long been on the roadmap for most airports. In addition to installing touchless fixtures, toilets, and dispensers, airport operators have looked for ways to present a cleaner environment. Recent efforts include enlarging the entryways to allow for physical distancing, improving air ventilation, and adding monolithic walls and floors (little-to-no seams) that make areas easier to clean and more clearly reflect that they are clean. Others include larger stalls with increased physical space as well as a "one way in, another way out" approach to eliminate bottlenecks and reduce interactions among passengers.

Some airport operators have added sensors to monitor restroom usage and alert staff when areas need to be cleaned as well as sensors that measure the levels in dispensers or trash cans. Additional modifications include gesture control door handles and dynamic signage outside the restroom to indicate occupancy levels.

Technology Readiness: Existing and emerging technology.

Barriers: There may be challenges opening the architecture of older facilities to add infrastructure.

Implementation Highlights: MSP increased the size of their entryways, improved air ventilation, and added monolithic walls and floors to give a clean look. **SEA** and **DFW** (*12*) have installed stall-use lights and digital screens at restroom entries that display restroom availability. **Flagship**, an airport maintenance service provider, is beginning to look at restroom throughput, stall indicator lights, and sensors for trash cans, paper towel, tissue, and soap dispensers. **MSP** is using an improved ventilation approach, so the air is not shared with air in the hallways. **DFW** (*13*) has added heating, ventilation, and air conditioning (HVAC) system upgrades and UV technology to improve the air quality within the facility. **PIT** installed sensors to count people using the restroom. They are using artificial intelligence (AI) to look at the optimal cleaning schedules. **YWG** is testing and considering an AI solution with video cameras to alert staff when a certain number of people have gone into the restrooms. These tools, also being considered by **Flagship,** can be used to build cleaning schedules based on actual facility usage instead of simply the passage of time.

Getting Started: Most airports are minimally installing touchless faucets, soap dispensers, paper towel dispensers, and hand dryers. If an airport is already planning for a remodel or new facility construction, the more extensive options noted here become much more feasible.

Enhanced Surface-Cleaning-Related Solutions

Antimicrobial Coatings	Cleaning Accreditation	Disinfectant Sprays	Escalator and People Mover Automatic Cleaning	Visible Cleaning Procedures
UV-C Autonomous Cleaning Robots	UV-C Footwear Sanitization		UV-C Pod for Wheelchairs and Trolleys	Other UV-C-Related Solutions

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Since there are still several surfaces that passengers and employees are required to touch throughout the airport campus, cleaning standards have been enhanced, and several technologies have been introduced to mitigate the effects of required touchpoints. Cleaning protocols vary greatly across airports, but all have the intention of reducing germs on high-touch surfaces. The solutions noted here can have a large impact on the efficiency and effectiveness of an airport's cleaning efforts.

Technology Readiness: Existing and emerging technology.

Barriers: Health guidance and regulations seem to be regularly evolving.

Implementation Highlights: LHR has added antimicrobial coatings across all key passenger touchpoints including door handles, handrails, self-service check-in kiosks, and bag-drop machines.

PHX (*14*) added seven UV-C light sterilizers to presecurity escalators' handrails. The sterilizers were placed out of sight underneath the escalators, so large signage was added to let passengers know the escalators' handrails were clean. The airport then put in plans to add the sterilizers to all escalators. **MSP** has started using UV-C light for ventilation, handrails, and bin return. TSA in **LAS** (*15*) is testing an integrated UV-C sanitization for the bins as they pass through an automated bin return system. **GRR** (*16*) tested several UV-C light use cases including pods to sanitize wheelchairs and baggage trolleys, footwear sanitization stations, and a personal item chamber.

EWR added a self-service bin cleaning station at security so passengers could clean bins themselves. The cleaning station included hand sanitizer, wipes, and a trash can.

GRR was one of the first airports to test the UV-C robot in baggage claim and the security checkpoint. Although it could disinfect a room without chemicals, they did not implement the technology beyond the pilot. **EYW** deployed their autonomous robot, R-Two-Key-Two, to clean the interior of the airport. **EWR** considered using autonomous robots but did not move forward with them due to cost. Instead of implementing newer technologies, **YWG** chose to shift their cleaning to be more visible to the passengers.

GRR, **SEA**, and **EWR** received the GBAC STAR facility accreditation. **Flagship** helped seven of their airports go through GBAC STAR facility accreditation. **DL** started Adenosine Triphosphate (ATP) testing and has a global cleanliness team. Over 400 airports have received the ACI Airport Health Accreditation for health and safety, such as **MSP**, **YWG**, **EWR**, **SEA**, and **PHX**.

Getting Started: Minimally, airports should seek to increase the frequency and visibility of existing cleaning procedures. Several airports have formalized their cleaning procedures and achieved accreditation to reassure passengers.

Passenger-Assistance-Related Solutions

Ambassadors	Social Media	Virtual Ambassadors	Augmented Reality Wayfinding	Wayfinding Beacons

These solutions help passengers navigate the airport, get real-time assistance, or provide feedback without having direct contact with an agent or staff member. See Appendix A for more information regarding how to assist travelers with disabilities at your airport.

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Technology Readiness: Existing and emerging technology.

Barriers: It is necessary to inform the public that these technologies are available. Passengers may be hesitant to use virtual information booths if they think they are bothering the assistant with a trivial question.

Implementation Highlights: SDF (*17*) opened a virtual information booth that allows passengers to connect to a live video feed of a volunteer airport ambassador (Figure 7). The staff member is onsite but located in another place in the airport to reduce direct interaction with the user. **LAX** (*18*) launched a pilot program in Terminal 2 that provided users access to a guest experience member via a touch-free tablet with a two-way video feed. **DEN** and **BWI** (*19*) also have virtual information desks. **UA** passengers at **ORD** and **IAH** (*20*) can scan a QR code to launch a virtual agent for real-time assistance over their mobile devices. **RDU** uses a guest experience team to help communicate changes or provide help in the terminal. They are working on a program that would include screens located throughout the airport similar to the other airports noted.

HKG and **MIA** (*21*) use their beacon-enabled apps to provide interactive navigation maps or trigger location-relevant messages. **SFO** (*22*) has a beacon-enabled app that also uses Apple's voiceover technology to give voice-based directions in real time.

LGW (*23*) and **ZRH** (*24*) have released wayfinding using an augmented reality view on a passenger's mobile device.

Getting Started: Passenger assistance solutions should start by identifying the most common needs and periodically emerging needs (such as those associated with traveling during a pandemic) of the passengers. Initial solutions can be low tech and progressively developed into touchless and seamless solutions. **MSP** uses volunteers to help share information and direct passengers to less-congested areas or shorter lines. **TSA** (*25*) launched "@AskTSA" on Twitter and Facebook Messenger to help passengers locate lost items and ask about prohibited items or screening policies in real time.



Figure 7. SDF Virtual Information Booth (photo by Louisville Muhammad Ali International Airport).

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Smart Building Systems

Lighting	Ventilation	Heating and Cooling

The use of smart building systems can regulate various components, including lighting and ventilation, without users needing to continually interact with controls. Lighting is controlled based on occupancy and outdoor natural light levels; ventilation is driven by CO_2 sensors and outdoor temperature to modulate indoor air requirements. While the adaptive nature of smart controls reduces touchpoints, the main drivers for these systems are energy efficiency, cost reductions, emissions reductions, and passenger comfort.

Technology Readiness: Existing and mature technology.

Barriers: Complexity of the spaces and passenger flows may be complicating factors.

Implementation Highlights: SEA implemented touchless lights in conference rooms and have daylight sensors control the lights in the terminal. **Flagship** is looking to add indoor air quality sensors.

Getting Started: This solution provides opportunities to improve energy efficiency and should therefore be reviewed as part of a holistic approach to building design and operations. Some standalone initiatives for specific rooms or areas could initially be deployed locally.

Signage

The onset of the pandemic drastically changed airport processes and how passengers and employees move through the airport campus. Signage has played a critical role in helping to communicate closures within the building, new processes, or room capacity. See Appendix B for more detail regarding the signage strategy at airports.

Technology Readiness: Existing and emerging technology.

Barriers: Static signage is easy and cost-effective to implement, but it takes effort and additional cost to reprint and hang static signage when conditions change. Dynamic signage makes it easier to implement real-time changes; however, there are additional upfront infrastructure costs and optional ongoing content costs.

Implementation Highlights: MCO (*26*) combined digital signage with crowd monitoring technology to provide a visual indicator of the current capacity of holdrooms (see Figure 8). Passengers can choose a less crowded place to sit based on the visual guidance. **PIT** changed digital signage by the trains to let passengers know when the next train is coming. Knowing that a train is coming shortly after the current train enables passengers to make the decision that they do not need to pack onto the current train. **MSP** used dynamic signage to communicate security wait times to help passengers choose the least congested checkpoint. **PIT** and **SEA** used light and humorous signage with a local-centric feel and flavor to communicate their messaging regarding new processes.

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Getting Started: Airports can add static signage on the floors, walls, and overhead. As signage is used in all areas of the airport, consider developing a signage plan to help identify priority areas and to establish a means of considering new signage points. However, be aware of and plan for any permanent impacts of signage installations to surfaces.





Figure 8. MCO Airport: Synect's Evenflow Crowd Radar and ReadySeeGo (photos by Synect).

Queue Management Systems

Behind the scenes, queue management systems allow airport operators to analyze their queue levels and view real-time data. These solutions allow for advanced data analysis that can be used to make changes in staffing levels, queue layouts, and airport schedules to reduce queue peaks. Additional features allow for real-time and predictive wait time metrics that can be shared with passengers via a website, an app, or dynamic signage. At the height of a pandemic, it is important to ensure the traveling public is maintaining a safe distance and areas are not overly populated. Queue management systems provide airport operators with the capabilities to manage crowd density, monitor social distancing metrics, and apply business rules to help with mask detection (as needed).

Technology Readiness: Existing and emerging technology.

Barriers: Queue management systems need cameras or sensors to capture the queue. The height of the ceiling in an area can often pose issues with the viewing angle or distance. In addition, there may be some difficulty configuring areas in the software if the queue lanes change often.

Implementation Highlights: GRR has a queue management system that can report the number of passengers in the queuing areas and measure the distance between passengers. They were able to use this data to better manage their queue lines, inform TSA of times when they should staff up, and inform airlines coming in of the more crowded times. **MSP** uses analytics to see where they can monitor and improve utilization to enhance the experience. They noted business intelligence and the ability to look ahead at flight schedules and passenger flows, ticket counters, and checkpoints. **MIA** (*27*) uses motion analytics technology to measure the distance between passengers and gather data about passenger movements while in the queue. **SEA** is in the process of integrating their queue management system with the new virtual queuing system.

Getting Started: Advertising waiting times based on manual time keeping or assessing the time between a boarding pass scan at the beginning and the end of the queue can be an initial solution for those unable to implement a full system.

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Honorable Mention Solutions

The following solutions may not be as valuable as those already mentioned, or they may already be ubiquitous across the industry.

- **Personal hygiene and health-related solutions:** In response to the pandemic, most airports began to add personal hygiene and health-related solutions around the terminal. For example, **EWR** added plexiglass across all concessions, check-in, and gate podiums to minimize the direct interaction between passengers and staff. In place of other solutions such as UV light or antimicrobial coatings, **PIT** positioned hand sanitizer stations on both sides of doors and near anywhere there would be food handling. In addition, they placed automatic hand washing stations outside the train (see Figure 9).
- **Touchless water bottle filling stations:** Water bottle filling stations support the importance of drinking water during air travel and promote sustainable behaviors. Touchless stations use sensor technology that is identical to that of water faucets used in restrooms and allows users to refill their bottles without touching the drinking fountain. These stations were especially important during the beginning of the pandemic when most drinking fountains were turned off for safety. Airports such as **SDF**, **SFO**, and **MKE** offer touchless water bottle filling stations.
- **Feedback devices:** Capturing passenger feedback within the terminal is a critical piece of assessing operational efficiency and customer satisfaction. **ATL** (*28*) trialed touchless feedback devices that allow the passenger to give a thumbs up or thumbs down gesture instead of pushing a button on a device. Other solutions employ a QR code that sends a user to a website on their mobile phone to give feedback.



Figure 9. PIT automatic hand washing station (photo by Pittsburgh International Airport).





Parking, Arrival, and Exit

Solutions that specifically apply to parking garages and lots, entry roadways, intermodal facilities, curbside, or other related areas.

Every airport has a unique series of roadways, parking areas, and intermodal facilities. The airport's parking and arrival and exit areas involve a transportation mode transfer, often from a private car. The focus in this area of the journey is on minimizing touchpoints and interactions

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with people to complete the intermodal transfer, whether through a prebooked car parking reservation and payment or prebooked public or private transit options. The following solutions can help make these areas more touchless.

Parking Reservation Systems

These systems enable users who have prebooked a parking reservation to present a QR code to gain entrance to a parking facility. Users are later able to exit the facility by scanning the same QR code, which allows them to bypass the payment kiosk or cashier. An emerging trend is combining the parking reservation system with license plate recognition (LPR) technology. This allows users to skip scanning a QR code upon exit. See the later section regarding LPR for more detail.

Technology Readiness: Existing and mature technology.

Barriers: Clear guidance for exception management is required in the case of a lost or unreadable QR code.

Implementation Highlights: MSP implemented this system to lower the number of parking kiosk touches and agent interactions. **PHX** and **SDF** began implementing the parking reservation system before COVID-19 to simplify operations in the parking facility, but during the initial months of the pandemic, they recognized many of these simplified operations were touchless enhancements; now, all facilities are included in the reservation system. Airports such as **RDU**, **PIT**, **SFO**, and **SEA** have also implemented these systems.

Getting Started: A first step is to have a conversation with your parking management firm and identify quick wins that could enable passengers to prepay parking and get a confirmation/code that provides seamless access once at the airport.

Parking Entry and Exit Kiosks

Many airports have already implemented ticket kiosks at the parking facility entrance and payment kiosks at the exit, which eliminate the need for passengers to interact with a cashier.

Technology Readiness: Existing and mature technology.

Barriers: Clear guidance for exception management is required in the case of a lost or unreadable ticket.

Implementation Highlights: GTR recently implemented this approach, noting that, in addition to the touchless benefits, they were also able to recover revenue.

Getting Started: A first step is to discuss with your parking management firm any quick wins that could enable passengers to pay for parking without seeing the staff.

Parking Pay-on-Foot Kiosks

Parking pay-on-foot kiosks allow customers to stop at a kiosk and pay for their parking fees before approaching the exit. Although many of these kiosks do not offer touchless or contactless payments yet, they do reduce interactions with the cashiers.

Technology Readiness: Existing and mature technology.

Barriers: Clear guidance for exception management is required in the case of a lost or unreadable ticket.

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26 Toward a Touchless Airport Journey

Implementation Highlights: Airports such as **SFO**, **RNO**, **SMF**, **ATL**, **SEA**, **and TUS** have implemented parking pay-on-foot kiosks. **RDU** customers have the option of contactless payments using their smart phone and watches.

Getting Started: A first step is to discuss with your parking management firm any quick wins that could enable passengers to pay for parking without seeing staff, such as through payment at a kiosk before exiting.

License Plate Recognition

LPR technology can be used to capture the customer's license plate on arrival and connect it to their reservation or parking ticket. Then, upon exit, LPR technology reads the plate, retrieves the customer record, and either (a) displays the fee total on the screen without requiring interaction or (b) automatically opens the exit gate (for users who have already provided payment information). Optionally, users can find their cars using a kiosk that uses LPR technology within the parking facility. This eliminates the need to call an attendant to help locate lost vehicles.

Technology Readiness: Existing and mature technology.

Barriers: Privacy and consumer protection regulations may be applicable. License plate covers or bad lighting may prevent the system from getting a good read.

Implementation Highlights: RDU (*29*) has implemented LPR that automatically displays the fee for those who need to make payment and opens the exit gate for those who used the parking reservation system. **SLC** (*30*) is using LPR to offer find-your-car functionality as an extension of the parking guidance system. Other uses for LPR include **ATL** (*31*) using LPR for inventory and audit purposes, and **DFW** (*32*) utilizing LPR to help prevent fraud. **YWG** is considering a pay-by-plate technology.

Getting Started: Airports will likely already have parking solutions implemented and should therefore reach out to their parking management provider to identify whether they have LPR functionality.

Virtual/Digital Queue Solutions

VQ solutions can be used to turn any physical queue into a digital queue. In this portion of the journey, QR codes can be posted in the parking facility to allow passengers to scan and join a VQ in real time as they head to the airport terminal. Another way for passengers to join the queue can be done through the integration of applications such as TNCs and parking systems with the VQ system.

A variation of the VQ is available for taxi drivers that allows drivers to join a digital queue without being in a physical queue. Very similar to the functionality provided for TNCs, this functionality alerts drivers when it is time to return to the airport for a pickup. This can reduce the need for taxi drivers to interact with a parking attendant, as well as reduce congestion in the staging area.

Technology Readiness: Existing and emerging technology.

Barriers: It is necessary to educate the public that this technology is available to reserve a spot in the taxi queue before arriving at the airport terminal.

Implementation Highlights: MSP deployed "eDispatch" in 2019 to allow drivers to join a VQ from their smartphones. **PANYNJ** is currently deploying a virtual dispatch

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system that eliminates the need for large holding areas at **JFK**, **LGA**, and **EWR**. **SFO** (*33*) utilizes the "TaxiQ" phone application which allows drivers head of the line queue privileges after a short trip.

Getting Started: As with the indoor VQ solution, airport operators could have drivers join the queue using a QR code located in specific portions of the airport.

Remote Baggage Drop

Remote baggage drop allows passengers to check baggage before they arrive at the main airport terminal. In this step of the journey, remote bag-drop locations allow passengers to streamline their check-in process or bypass the check-in lobby altogether.

Technology Readiness: Existing and mature technology.

Barriers: There are regulatory requirements for bag acceptance, bonding, and screening.

Implementation Highlights: TPA (*34*) allows passengers to check their bags at the rental car center so they can skip the queues at the main terminal. Airports such as **BOS** (*35*), **LAS**, **SFO**, and **PHX** have also implemented remote bag drop.

Getting Started: This solution should be reviewed as part of a broader assessment of check-in demand and capacity to highlight locations in car parking and drop-off areas that would specifically benefit from a bag drop-off facility.

Self-Service Kiosks

Mobile Screen Control	Key Fobs	Hover Screen Control

Self-service kiosks allow users to complete functions without the direct assistance of an employee or agent. In this portion of the journey, self-service check-in kiosks can be used to allow passengers to check-in for their flight, purchase seat upgrades, change seat assignments, and print baggage tags (see the "Landside" section for additional technology details). These kiosks are typically located in the main airport terminal; however, many airports have kiosks located in the rental car center, parking garage, or other remote locations.

Technology Readiness: Existing and mature technology.

Barriers: Regulatory requirements related to identity checks, bag tag activation, and travel health checks may impact kiosk features or subsequent processes.

Implementation Highlights: MCO has kiosks located in their intermodal facility to help passengers prepare for their journey before arriving at the main terminal.

Getting Started: This solution should be reviewed as part of a broader assessment of check-in demand and capacity to highlight locations in car parking and drop-off areas that would specifically benefit from the deployment of check-in kiosks. Also, consider any potential issues in placing these kiosks outdoors in the humidity and exposure to rain.

Automated Car Park Systems

Airport operators are testing several applications of automated car park systems for passenger vehicles and rental car fleets. The potential of this solution is growing as AI and automation are enhanced.

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Technology Readiness: Existing and emerging technology.

Barriers: Barriers exist in the areas of cost-benefit, technology readiness, and buy-in from users.

Implementation Highlights: Airports such as **CDG**, **LYS**, **LGW** (*36*), **and DUS** (*37*) are using automated car valets that use autonomous robot valets to park cars. **STR** (*38*) is testing another variation of the automated car park system that will allow drivers to use their smartphone to tell their car where to park.

Getting Started: These systems are complex and costly. Airports should initially identify a specific application and location where such a system would solve operational issues or maximize the use of a constrained area.

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport.





Landside

Solutions that specifically apply to the check-in lobby, landside concessions and shopping areas, and baggage claim.

Landside relates to all processes, touchpoints, and discretionary actions that a passenger and staff can undertake prior to security screening or after arriving on a flight and clearing relevant border processes if applicable. The following solutions apply specifically to this area of the journey.

Autonomous Wheelchairs

Autonomous wheelchairs help those with reduced mobility get through the airport without close contact with an attendant and without the wait for an available attendant. These wheelchairs have anti-collision capabilities using sensor technology and automatic brakes as they navigate through the building. After delivering passengers to their destination, the wheelchair drives itself back to the docking station.

Technology Readiness: Emerging technology.

Barriers: Regulations around automation evolve as trials are pursued. There are also navigation issues in crowded concourses and multilevel facilities.

Implementation Highlights: YWG has piloted autonomous wheelchairs in the parking lot, at the curb, and at the check-in counter. Other pilots include **TPA** (*39*), **DFW** (*40*), **JFK** (*41*), **HND** (*42*), and **AUH** (*43*).

Getting Started: Existing wheelchair operations should be reviewed to identify applications or locations where the implementation of an autonomous wheelchair could be considered.

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Self-Service Kiosks

Mobile Screen Control	Key Fobs	Hover Screen Control

Self-service kiosks allow users to complete functions without the direct assistance of an employee or agent. In this portion of the journey, self-service kiosks can allow passengers to check-in for their flight, purchase seat upgrades, change seat assignments, and print baggage tags. In response to the pandemic, several solutions have emerged to reduce passenger interaction with the kiosk, including hover technology or mobile control using either Bluetooth or a QR code. Separate from check-in kiosks, quick bag tag printing kiosks provide another option to quickly print bag tags.

Technology Readiness: Existing and emerging technology.

Barriers: Regulatory requirements related to identity checks, bag tag activation, and travel health checks may impact kiosk features or subsequent processes. Passenger adoption varied greatly across airports.

Implementation Highlights: PIT check-in kiosks display a QR code that passengers can scan to (a) select options on the screen or (b) drive a mouse pointer. **MSP** is exploring the touchless kiosk solution, which utilizes a QR code and allows the passenger to complete all transactions on their phone. **EWR** is looking into using the mobile phone as a mouse on the kiosk. **RDU** is exploring adding touchless capabilities on their kiosks directly using the airline apps. **SEA** piloted a "happyhover" overlay on the kiosk that allows the passenger to select options available at the kiosks without direct touch; however, they noted that passenger adoption was low. Similarly, **HKG** added proximity sensors for check-in kiosks that allowed passengers to interact with the screen by hovering over it without touching it.

Getting Started: Airports can provide low-cost key fob or stylus tools to give passengers the option to touch the screen with a separate object. Another obvious solution is to place hand sanitizer stations near the kiosk. Additional options and features should be reviewed with manufacturers and should focus on solutions that can easily be implemented and adopted by passengers.

Self-Service Bag Drop

Self-service bag-drop systems allow passengers to induct baggage into the baggage handling system without direct interaction with an agent. These units perform the following functions: (a) verify the passenger's identity, (b) print the bag tag(s), (c) measure and weigh the bag, (d) optionally accept payment for extra or overweight baggage, and (e) accept the bag into the baggage handling system. Recent pilot projects have incorporated the use of facial biometrics and the TSA's database to verify the passenger's identity to make it a touchless, seamless experience. Self-service bag-drop systems can be installed in a one-step (all passenger functions at the unit) or two-step (check-in and bag tagging at a self-service kiosk first) fashion, depending on airport and airline preferences.

Technology Readiness: Existing and emerging technology.

Barriers: There are barriers in regulatory requirements around baggage acceptance and use of biometrics as well as the need for user buy-in.

Implementation Highlights: ATL (*44*) features a biometric, touchless, end-to-end experience that starts in a dedicated space called the Delta-TSA PreCheck Express Lobby and Bag Drop (see Figure 10). Passengers who check in on the Delta mobile app and who

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are members of Delta's SkyMiles program and TSA PreCheck can use their face to complete the self-service bag-drop process. Other airports with self-service bag-drop units include **DEN** (*45*), **HKG** (*46*), **LGA** (*47*), **LHR** (*48*), **ORD** (*49*), **LAS** (*50*), **HND** (*51*), and **SIN** (*52*). Previous self-bag-drop pilots have been completed at **MSP** (*53*) and **LAX** (*54*).

Getting Started: Bag-drop units should be assessed as part of a broader check-in strategy. Some models can be retrofitted into existing counters to minimize infrastructure requirements.





Figure 10. ATL self-service bag drop (photos by Delta Air Lines).

Virtual/Digital Queue Solutions

VQ solutions can be used to turn any physical queue into a digital queue. In this landside portion of the journey, QR codes can be posted around the terminal to allow passengers to scan and join any VQ that the airport is offering. Although the recent VQ uses in airports have focused on security, it could be very beneficial to use VQ for landside queues such as those in the check-in hall.

Technology Readiness: Existing and emerging technology.

Barriers: It is necessary to inform the public that this technology is available. VQ technology is not new; however, its usage in airports is new. Currently, the total cost of this solution is still unknown.

Implementation Highlights: Passengers who wanted to use the VQ pilot in **BOS** scanned QR codes located near the airport entrance. In addition to online reservations, passengers can scan onsite QR codes to join the VQ in real time at **SEA** and **LAX**.

Getting Started: Alternate methods to join the VQ include providing onsite kiosks or posting a phone number so users can text to join the queue.

Biometric Technology

Biometric technology solutions can be added to self-service kiosks and self-bag-drop units to automate identity verification during the passenger journey. Biometrics can be obtained from a passenger's face, eyes, or finger; however, the facial biometric is most often used in the United States. In this portion of the journey, if the passenger has not yet enrolled their biometrics into the program, the passenger would do so before their biometrics could be validated. More recently, biometrics are also being discussed as an addition to VQ solutions.

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Technology Readiness: Existing and emerging technology

Barriers: There are barriers in privacy regulations, the need for user buy-in, and a seamless opt-in process. See the "Pre-Airport" section of this guide for more detail.

Implementation Highlights: HKG currently has facial recognition enrollment at the check-in kiosks, which is stored as the passenger "single token." The **IATA** One ID uses your face as your identity, looking at biometrics to remove the need to stop at every point to show documents. IATA is trying to design how this technology will be used to improve the passenger process, not only for airline touchpoints but also for border patrol and security. **DL** has launched the first fully biometric terminal, which offers a curb-to-gate biometric experience for international customers. **SFO** (*55*) is trialing a biometric-enabled airport experience in which passengers link their driver's license to their facial biometric at check-in. The passenger then scans their face to complete check-in, bag drop, and boarding.

Getting Started: The use of biometrics should be assessed across the entire passenger journey. Standalone applications using the ePassport may initially be easier to implement as its use does not require pre-enrollment.

Automated Access Systems

Electronic gates (eGates) are automated barriers that allow passengers to manually scan a boarding pass or undergo biometric verification to gain access to another area (see the "Security" section for additional technology details.) eGates have typically been utilized at boarding gates in the United States and are beginning to see wider adoption in other portions of the airport.

Technology Readiness: Existing and emerging technology.

Barriers: Gates must demonstrate an equivalent level of compliance to a manual check.

Implementation Highlights: TPA (*56*) recently added eGates to ensure passengers board the correct shuttle to the airside (see Figure 11). Passengers scan their boarding pass; the system checks to make sure they are going to the correct airside, then they are granted access to board the shuttle.

Getting Started: Regulatory requirements should be reviewed in detail to ensure the eGates specifications meet the required outcome.



Figure 11. TPA eGates (photos by EASIER).

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Honorable Mention Solutions

Honorable mention should be made for the following solutions in this area of the journey:

- Spirit Airlines' **virtual assistance** implementation of the baggage service office**:** Instead of staffing agents in a small baggage service office, Spirit has added QR codes on signage and allowed passengers to interact with virtual agents to make a claim for lost baggage.
- **Retail, food and beverage purchase, and delivery options** should also be considered for this portion of the journey (see the "Airside" section for additional technology details).
- **Access control systems** help secure sensitive areas of the airport from the traveling public (see the "Back of House" section for additional technology details).

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport.





Back of House

Solutions that apply to staff-specific areas such as baggage processing and airport management offices.

Back of house is a staff-specific environment with workplace measures like those available to passengers to minimize touchpoints.

Access Control Systems

Access control systems help secure sensitive areas of the airport from the traveling public. These systems traditionally allow access using a badge, pin code, or both. Conventional access control is being improved by enhanced biometric features for a more touchless experience.

Technology Readiness: Existing and emerging technology.

Barriers: Access control to sterile or restricted areas is regulated. The solution must meet regulatory requirements. There are also privacy regulations around biometrics.

Implementation Highlights: All airports deploy access control systems. Interviews did not draw out specific insights on how these airports are moving to touchless operations. Some solution providers offer technology that marries video sensors with an AI algorithm to quickly recognize the face before a user approaches a door. The **Canadian Air Transport Security Authority** (*57*) uses Restricted Area Identity Cards with dual biometric access control (iris and fingerprint).

Getting Started: Access control is driven by regulations applicable to the airport and must be minimally met.

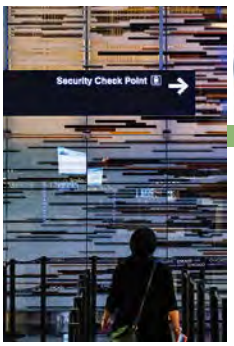
Honorable Mention Solution

Honorable mention should be made for checked **baggage screening systems**, specifically computerized axial tomography (CT) x-ray baggage screening systems that provide 3D images that can be rotated on three axes for a complete image analysis. The unit analyzes the 3D images using

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sophisticated algorithms that can detect explosives, weapons, and other prohibited items in checked baggage. The CT scanners reduce the need for baggage to be manually checked by a TSA agent.

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport. Examples for back of house include restrooms, elevators, surface cleaning, automated water bottle filling stations, and smart building systems.



Security

Solutions that specifically apply to the security screening areas managed by the Transportation Security Administration.

The security screening process is historically a manual process with several steps of staff interaction. Increasingly, technology and process enhancements are enabling reduced touchpoints and interactions.

Virtual/Digital Queue Solutions

VQ solutions can be used to turn any physical queue into a digital queue. Users can join the queue before arriving or once they have arrived at the airport terminal (see the "Pre-Airport" and "Landside" sections). In this area of the journey, passengers present their QR code to enter the screening area at their designated time. This allows for an expedited process and reduces the amount of time the passenger stands in the physical queue. A VQ can improve the customer experience, help reduce congestion, and ultimately be used to assist with load distribution during busy periods. VQ can also indirectly increase revenue by giving passengers more time to shop as well as reduce their level of stress.

Technology Readiness: Existing and emerging technology.

Barriers: There is a complexity of process to getting through security when it is a passenger's turn. There also need to be sufficient presecurity dwell areas.

Implementation Highlights: SEA (*58*) has had great success with the Spot Saver VQ and moved it from pilot to full deployment in September 2021 (see Figure 12). Sixty-three percent of surveyed passengers using SEA Spot Saver said they were able to relax or purchase something because they were not standing in line. **BOS** and **LAX** (*59*) are additional trial locations. Based on conversations with BOS and SEA, the pilots in 2021 showed great opportunity for VQ to flatten or reduce key peaks at screening checkpoints, encouraging infrequent or nervous travelers who would usually arrive 3 to 4 hours early—increasing congestion at the busiest hours—to feel more comfortable having a reservation and allowing the airport to suggest the time they arrive.

Health-conscious passengers who opted into the VerfiFLY VQ pilot in **DEN** (*60*) accessed a separate screening lane and limited-capacity tram to travel to their gate. **PIT** is working on a mobile platform that would include digital queuing technology. Although it is currently on hold due to the pandemic, **YUL** (*61*) was the first airport to offer VQ functionality, called SecurXpress, to passengers in 2014.

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Getting Started: VQ solutions at the checkpoint involved the airport, airline, and TSA. To get started, discuss the options and desires of your stakeholders. To save money, use staff to assist with the VQ-to-physical queue entry instead of fully automated eGates. According to VQ pilot program participants, gathering a baseline of the airport operator's and TSA's current passenger wait times, queue lengths, and other key metrics is very helpful. An assessment will be needed to understand how many passengers per time or slot can be managed in the VQ without significant impacts on general screening queues.

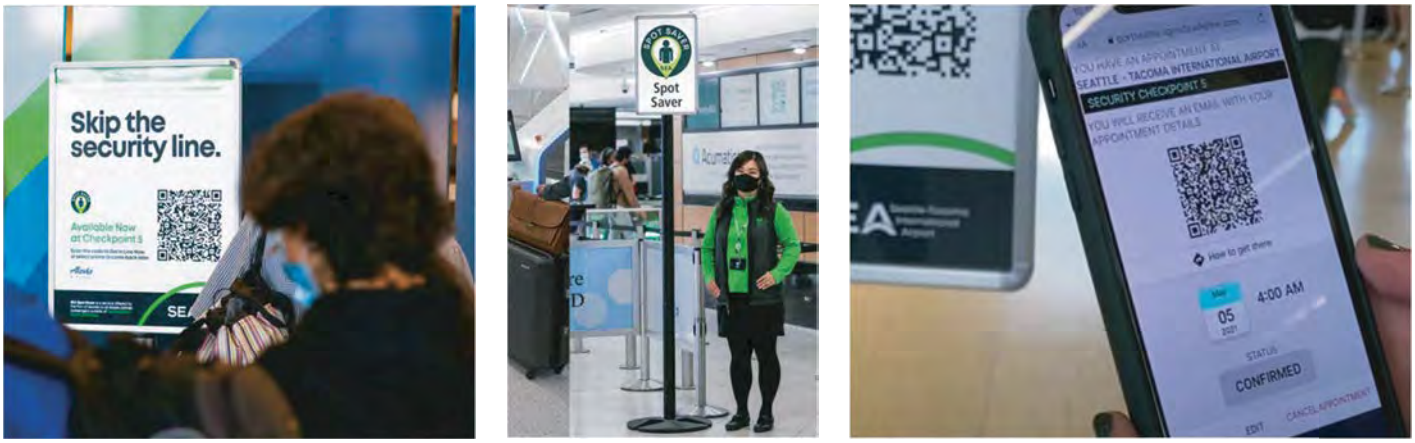


Figure 12. SEA Spot Saver (photos courtesy of Port of Seattle/Seattle-Tacoma International Airport).

TSA Credential Authentication Technology

Credential Authentication Technology (CAT) verifies identification documents (driver's license or passport) presented at the TSA security checkpoint. The CAT machine is linked electronically to TSA's Secure Flight database. This confirms travelers' flight details, ensuring they are ticketed for travel that day. CAT also displays any eligible prescreening status such as TSA PreCheck. These machines eliminate the need for passengers to hand a boarding pass to a TSA agent. In many cases, the CAT machine is turned to face the passenger so they can directly insert their driver's license, eliminating the need for the agent to handle the document. A recent enhancement pairs the CAT with a camera to validate that the identity presented to the CAT machine matches the person by capturing a live facial image, providing a biometric one-to-one match.

Technology Readiness: Existing and emerging technology.

Barriers: Barriers may include ID check regulations as well as the additional infrastructure that is needed at the podium location.

Implementation Highlights: CAT (*62*) is now implemented at over 120 U.S. airports including **GRR**, **MSP**, **PIT**, **RDU**, **SDF**, and **SEA**. Biometric one-to-one matching with CAT has been trialed at **LAS** (*63*) in 2019 and more recently at **DCA** (*64*).

Getting Started: Airport operators should begin talks with the TSA or join TSA's Innovation Task Force (see https://www.tsa.gov/itf for more information) to receive new features more quickly.

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Baggage Screening Systems

Like the larger CT x-ray baggage screening systems, smaller versions of the CT are used at the TSA checkpoint to screen carry-on baggage. The systems provide 3D images that can be rotated on three axes for complete image analysis. The unit analyzes the 3D images using algorithms that can detect explosives, weapons, and other prohibited items in carry-on baggage. These CT systems allow passengers to keep laptops and electronics in bags, which reduces the direct handling by the TSA agent and expedites the screening process.

Technology Readiness: Existing and mature technology.

Barriers: Preboard screening is regulated. The solution must meet regulatory requirements.

Implementation Highlights: Over 130 airports have implemented CT systems at passenger security checkpoints, including **BOS**, **BWI**, **CVG**, **HOU**, **IAD**, **IND**, **JFK**, **LAS**, **LAX**, **MIA**, **OAK**, **ORD**, **PHL**, **PHX**, **SAN**, and **STL**.

Getting Started: Airport operators should begin talks with the TSA or join the Innovation Task Force to receive new features more quickly. Also, be aware of the space requirements of newer technology and reference the TSA Checkpoint Requirements and Planning Guide for details on the larger CT machines and the future Checkpoint Property Screening System.

Automated Access Systems

Electronic gates (eGates) are automated barriers that allow passengers to manually scan a boarding pass or undergo biometric verification to gain access to another area. In this portion of the journey, eGates can be used to verify identity and ensure the passenger is eligible to fly, and they can be deployed using a one-step or two-step process. Currently, this is primarily available for international travel. In the one-step process, the biometric is captured and identity is verified by the TSA or CBP, and the passenger is checked using the airline's departure control system (DCS) and allowed to pass. In the two-step process, the biometric is captured and identity is verified by the TSA or CBP. The passenger then scans the boarding pass, is checked using the airline's DCS, and allowed to pass.

Technology Readiness: Emerging technology.

Barriers: eGates need to demonstrate an equivalent level of compliance to a manual check.

Implementation Highlights: **HKG** implemented biometric eGates at the security checkpoint in 2018. Passengers can validate their facial biometrics compared to a previously created "single token" to gain entry. Alternatively, passengers can have their picture taken to capture their facial biometrics to create a single token at that time and gain entry. TSA at **DTW**, in cooperation with CBP and **DL** (*65*), began a round of testing facial comparison. This used the CBP's TVS to enhance passenger identity verification at the TSA checkpoint. Participation in the pilot is limited to SkyMiles passengers who use Delta's mobile app check-in process and who participate in the Trusted Travelers program. A similar pilot started in **ATL** (*66*) in October 2021 and features facial biometric checks at check-in, security, and boarding. **NRT** and **HND** (*67*) use facial recognition to allow international travelers to check baggage and pass security without showing passports or tickets. Passengers at **YEG** (*68*) and **BRS** scan their boarding pass for entry.

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Getting Started: Some implementations are achieved with a camera mounted on a pole. Though it does not include a physical gate, it can be an effective way to verify passengers.

Honorable Mention Solution

Honorable mention should be made for **Enhanced Advanced Imaging Technology** (eAIT), which screens passengers for prohibited items such as weapons and explosives without physical contact with a TSA agent. The open design body scanner does not produce an x-ray image but has updated algorithms that can detect prohibited items under the passengers' clothing. This reduces the passengers' interactions with agents and expedites the screening process. **DEN** (*69*) was the first U.S. airport to operate eAIT. **LAS** (*70*) utilizes eAIT at their Innovation Checkpoint.

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport.



Immigration Control

Solutions that specifically apply to the immigration and passport control area managed by U.S. Customs and Border Protection.

The immigration control process is increasingly becoming touchless with travel checks occurring upstream using biometrics to confirm identity. The following solutions apply to this area.

Automated Access Systems

In this portion of the journey, eGates can be used to verify identity using different applications of biometrics including ePassports, single token, or digital identity.

Technology Readiness: Existing and emerging technology.

Barriers: Baseline requirements are driven by regulations, regardless of the technology used.

Implementation Highlights: SIN (*71*) utilizes a single token biometrics process to compare the passenger's passport information and fingerprint biometrics. Passengers in **YWG** and **YYZ** (*72*) provide a facial biometric that is compared to the information on their scanned NEXUS card. Passengers from the United States can now provide facial biometrics and scan their passport to gain entry at 13 airports in the UK including **LGW**, **LHR**, and **MAN** (*73*).

Getting Started: Some implementations are achieved with a camera mounted on a pole. Though it does not include a physical gate, it can be an effective way to verify passengers.

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Self-Service Kiosks

Self-service kiosks allow users to complete functions without the direct assistance of an employee or agent. During this portion of the journey, passengers can use Global Entry self-service kiosks, which allow eligible passengers to scan their passport, submit fingerprints or facial biometrics, and complete their customs declaration. Although these kiosks are not completely touchless, they do reduce the amount of direct exposure to CBP agents.

Technology Readiness: Existing technology.

Barriers: Passengers may be reluctant to have their fingerprint or picture taken and decide to bypass this self-service option. Airports with spatial constraints may not be able to accommodate the kiosks and queues.

Implementation Highlights: Global Entry (*74*) kiosks can be found in over 70 international airports including **MSP**, **EWR**, **PIT**, **RDU**, **SEA**, **LAX**, and **YWG**.

Getting Started: Airports can provide low-cost, no-touch stylus tools to reduce touchpoints. Additional options and features should be reviewed with the manufacturer but should focus on solutions that can easily be implemented and adopted by passengers.

Biometric Technology

Biometric technology solutions can be used to automate identity verification during the passenger journey. Biometrics can be obtained from a passenger's face, eyes, or finger; however, the facial biometric is most often used in the United States. In this portion of the journey, the passenger can use their biometrics to utilize CBP's Simplified Arrival upon entry from international travel. This allows arriving passengers to submit their facial biometrics for identity validation. If matched to an existing passport or visa photo, the passenger can bypass the traditional manual document processing.

Technology Readiness: Existing technology.

Barriers: In this portion of the journey, barriers include not having a photo available to compare to the in-person photo as well as privacy regulations, the need for user buy-in, and a seamless opt-in or opt-out process.

Implementation Highlights: Simplified Arrival is deployed in **LAX** (*75*), **MIA** (*76*), **TPA** (*77*), **CLT** (*78*), and **KCI** (*79*).

Getting Started: The use of biometrics should be assessed across the entire passenger journey. Standalone immigration applications using the ePassport may initially be easier to implement as its use does not require pre-enrollment.

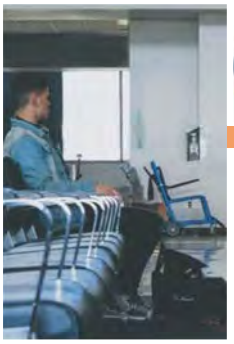
Honorable Mention Solutions

Honorable mention should be made for the following solutions in this area of the journey:

- **APC kiosks** allow arriving passengers to expedite the customs entry process. These kiosks are currently being replaced with the biometric-enabled CBP Simplified Arrival process (see the immediately previous "Biometric Technology" solution in this section for additional technical details).
- **Retail, food and beverage purchase, and delivery options** should also be considered for this portion of the journey (see the "Airside" section for additional technology details).
- **VQ** may be an additional technology utilized in this area to reduce congestion and improve the customer experience (see the "Security" section for additional technology details).

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See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport.





Airside

Solutions that specifically apply to areas such as holdrooms, airside concessions and shopping, and other amenities.

Airside relates to all processes or touchpoints that passengers and staff members encounter after completing security screening or before completing border clearing processes upon arrival on a flight. The following solutions apply to this area.

Automated Access Systems

In this portion of the journey, eGates can be used to verify passenger identity and allow access to the boarding bridge. See the description provided in the "Security" section of the journey for more information on this technology.

Technology Readiness: Existing and emerging technology.

Barriers: Baseline requirements are driven by regulations, regardless of the technology used.

Implementation Highlights: **MCO** (*80*) was the first airport in the United States to utilize eGates for biometric exit for all international flights. **LAX** (*81*) recently deployed facial biometric eGates for boarding (see Figure 13). JetBlue released a fully integrated, biometric, self-boarding eGate at **JFK** (*82*) after successful trials at **JFK**, **BOS**, and **FLL** (*83*). Passengers in **SEA** (*84*), **ATL** (*85*), and **DCA** (*86*) can board using only their facial biometric using a camera on a stick.

Getting Started: Some implementations are achieved with a camera mounted on a pole. Though it does not include a physical gate, it can be an effective way to verify passengers.



Figure 13. LAX biometric eGates (photo by Los Angeles World Airports, ©2020 Joshua Sudock).

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Food Ordering and Delivery

Contactless Payment	Self-Service Kiosks	Self-Service Tablets	Mobile App	Online Purchase (Pickup or Delivery)	Delivery Robots

Although not fully touchless, food ordering kiosks or tablets enable passengers to bypass the long lines and order food without staff interactions. Since the pandemic, food delivery options have greatly increased in popularity. Many airports are giving passengers an option to order from restaurants using a QR code, website, or mobile device. For passengers wanting to maintain physical distancing, airports are trying alternate delivery methods such as pick up lockers or robotic delivery systems.

Technology Readiness: Existing and emerging technology.

Barriers: It is necessary to inform the public that this technology is available. Another barrier is changing concessionaires' traditional methods from counter operations to meet the ever-evolving passenger requirements.

Implementation Highlights: **DFW** (*87*) and **MSP** airports have deployed self-service kiosks for food ordering. **MSP** has noted that they have seen approximately a 30 percent increase in ordering totals when using a kiosk versus face-to-face purchases. Airports such as **MSP**, **SEA**, **LAX**, **PHL**, **MDW**, and **SAN** (*88*) offer mobile food ordering and delivery via a QR code, app, or website. Figure 14 shows a food delivery robot at **SEA**. **EWR** (*89*) noted they can offer more concessions options and deliver food to satellite terminals where concessions were not yet open. **MSP**, **SEA**, **LAX**, **PHL**, **MDW**, and **SAN** (*90*) are also on the list of airports piloting contactless ordering systems featuring robotic food delivery. **YYZ** and **HMSHost** (*91*) were the first to launch a mobile order and pick up service in the airport using Uber Eats.

RDU is launching a virtual food hall or "ghost kitchen," which allows restaurants to operate from a central location and deliver food to a locker. This new concept will increase customer options without taking up additional concession space. **SAT** (*92*) just launched an automated food kiosk that prepares pizza. **CLT** (*93*) is one of the first airports to offer a smart self-checkout kiosk. This kiosk contains items behind a door that can be opened by scanning a QR code or swiping a credit card; a camera then tracks the customer's movements to charge for items removed from the kiosk. Once the door is closed, a UV light is activated to sanitize the remaining items.

Airport operators are also looking to automate retail purchases. In addition to pre-order options on mobile devices or websites, other emerging options include retail kiosks, queueless checkout options, and contactless payments. **Hudson** is now offering automated retail kiosks at **MYR** (*94*), **MDW** (*95*), and **SJC**. **Amazon's** Just Walk Out technology can be found at **DAL** (*96*) and **MDW** (*97*). Touchless self-checkout and payment are available at **KCI** (*98*), **CLT** (*99*), and **LAX**. This option uses AI to detect the items and display a price without human intervention.

Getting Started: Most airports agree that the use of self-service kiosks can increase revenue. Airports have also shown good adoption of food delivery services; automation can come later.

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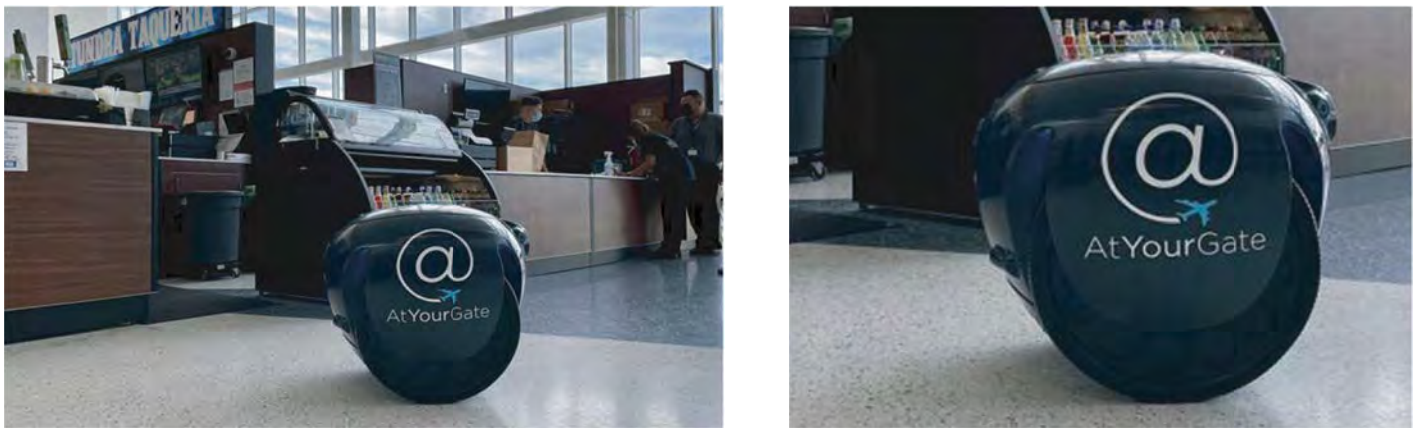


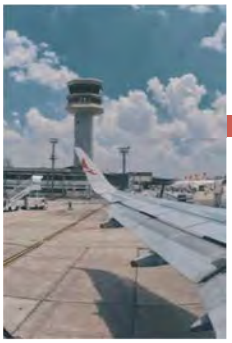
Figure 14. SEA robots for food delivery (photos courtesy of Port of Seattle/Seattle-Tacoma International Airport).

Honorable Mention Solutions

Honorable mention should be made for the following solutions in this area of the journey:

- **Access control systems** help secure sensitive areas of the airport from the traveling public (see "Back of House" section below for additional technology details).
- **Self-service kiosks** can be used to allow passengers to change their flight, purchase seat upgrades, change seat assignments, and print additional boarding passes (see the "Landside" section for additional details on this technology).

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport. Examples for airside include restrooms, queue management systems, passenger assistance, signage, elevators, surface cleaning, automated water bottle filling stations, and smart building systems.





Airfield

Solutions that specifically apply to airfield processes such as baggage handling and ground support.

The airfield primarily relates to the work environment for staff; however, it may be used by passengers being bused to another airside location or needing to walk to their gates. Though the research did not uncover significant solution implementations in this area, the solutions noted below are worth an honorable mention.

Honorable Mention Solutions

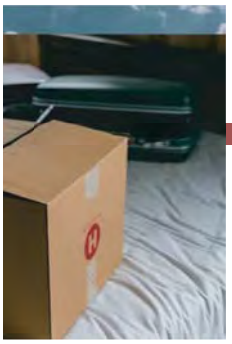
Honorable mention should be made for the following solutions in this area of the journey:

- **Access control systems** help secure sensitive areas of the airport from the traveling public (see the "Back of House" section for additional technology details).

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- **MSP** implementation of an automated solution for Part 139 Airport Certification inspections required by the Federal Aviation Administration.
- **Self-sanitizing gloves** provide an innovative approach to eliminating the spread of contaminates in operating areas. The gloves spray disinfectant on themselves and the surface upon contact.

See the "On-Airport" section of this guide for additional solutions that may apply to this area, depending on the specifics of your airport.





Post-Airport

Solutions that can move processes that typically occur on-airport to a post-airport location.

The post-airport environment benefits from the steps taken upstream to reduce the number of required touchpoints on the airport property. Though the research did not uncover significant solution implementations in this area, there are potential solutions regarding meeting the destination's travel or health document requirements, receiving baggage delivery, or delivery of goods and services ordered before getting to the passenger's destination.

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CHAPTER 3

Taking Action

Chapter 2 gave detailed descriptions of the solutions applicable to the areas along the airport journey. Now you can use this section to help decide which solutions to consider and implement for your airport. In addition, a downloadable solution worksheet, "Taking Action," is available at www.nap.edu by searching for *ACRP Research Report 241: Toward a Touchless Airport Journey*. The worksheet will walk you through your airport journey areas in the same way as Chapter 2 of this guide, enabling you to do the following:

- Check off the solutions you have already implemented.
- Identify which solutions you will consider next and take note of your next actions for each.
- Review the overall cost and IT requirements for each solution.

In building your worksheet, use the best practices and lessons learned identified in Chapter 1 of this guide:



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APPENDIX A

Helping the Passenger

Chapter 1 of this guide described six best practices: Connect with Airport Strategy or Technology Direction, Understand the Importance of Strong Leadership, Collaborate with Your Stakeholders, Focus on Helping the Passenger, Communicate with the Passenger, Leverage Additional Resources in the Community and Beyond, and Understand the Importance of Data.

The following section provides additional detail on "Focus on Helping the Passenger." This information is not necessarily specific to touchless solutions; however, these considerations are highly relevant as you consider the implementation of touchless solutions

This section identifies the typical barriers a passenger with disabilities or an older adult may experience throughout their passenger airport journey. This section details the accessibility issue followed by recommendations for mitigation. The information below is provided to enhance the touchless environment effort, which can be useful for all passengers traveling through the airport. For additional information on The Americans with Disabilities Act (ADA) visit www.ada.gov.



Pre-Arrival

Considerations

Passengers with disabilities often face challenges while preparing to arrive at the airport, including:

- Passengers may not know about assistance programs available at the airport.
- Contact information may not be available or easy to find on the airport website and/or mobile app.
- Passengers provide airlines with information about their need for assistance during their booking process; however, the information provided to the airport operator is limited.
- Airport maps often do not include the roadway, entryway to parking, curbside drop-off, or TNC drop-off points.

Mitigation Measures

- Provide education to the public that an assistance program, such as an identification program, is available.
- Include contact information on the website and mobile app in an accessible manner for all forms of disabilities. For example, consider requirements for those who are blind or have low vision.

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- Establish clear procedures for airlines to notify airport operators that they have a passenger who needs assistance. Coordinate with the traveler on expected arrival and/or offer a phone number that the passenger can call when they arrive.
- Ensure that airport maps are provided on the website and mobile app that include roadways, trains, shuttles (such as hotel and rental cars), TNC drop-off points, train station drop-off, or bus drop-off areas. The airport map should include a clear pathway to the terminal entryway, including specially marked curbside entryways for passengers needing assistance.



Arrival at the Airport

Considerations

Passengers with disabilities often face challenges while arriving at the airport, including:

- Lack of a designated entryway with a method for the passenger to communicate with guest services.
  - Issues arise because of a lack of awareness by the provider. An example is a person who is blind or has low vision and needs assistance from the door of the airport. Often there is no one there to greet them and help them get to the check-in counter or kiosk.
  - Some airports have courtesy assistance telephones or buttons that advise the passenger is there and needs assistance.
- Lack of clear wayfinding signage for those needing assistance. Passengers who arrive at the terminal must be assured of clear wayfinding signage when arriving at the entry point of the terminal.
- Passengers should be educated via the airport website or mobile app so they know where to go for services such as automated wheelchairs or identification lanyards.
  - If automated wheelchairs are available, that information needs to be posted on the website, along with a contact number to help the passenger identify where to park and get help if needed.
  - Passengers with cognitive disabilities may inform the airport customer experience of their disabilities so that they may be identified upon arrival and assisted if needed.

Mitigation Measures

- Ensure that airport staff, airlines, volunteers, third-party providers, and business partners are aware of designated entry points.
  - Establish procedures to ensure communication devices are in good working condition.
  - If communication devices are available, appropriate signage and/or auditory signal must be clearly identified.
- Work with an ADA coordinator and a community/airport committee to ensure that signage is clear for all types of disabilities and on all points of access. If you do not have an ADA coordinator, you can locate one in your region (https://adata.org/find-your-region). Standards for accessible design can be found at www.access-board.gov.
- Provide education to the public that services such as automated wheelchairs or identification lanyards are available.
  - Designate a specific point of entry for the individual traveling who has cognitive disabilities to obtain a lanyard.
  - Review the best location for posting signage for automated wheelchairs.

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- Airport operators should coordinate with airlines, volunteers, airport staff, and third-party providers to ensure that they know where automated wheelchairs are located, charged, and returned.



Landside

Considerations

Passengers may face challenges in the landside area including:

- Needing assistance getting to the check-in area.
  - If the passenger needs assistance to get to the check-in hall, the person assisting should meet the passenger at the designated entryway door of the terminal.
  - The kiosk(s) should be ADA accessible at a minimum, with auditory capability for the passenger who is blind or has low vision.
  - Gate information should be provided both in writing and verbally, so the passenger knows where to go.
  - Frequently, the wheelchair provider insists that the person sits in the wheelchair although the person with vision impairment is mobile, or they will not convey them to the gate.
- Needing assistance locating the following facilities or amenities:
  - Escalator and elevator access.
  - Restroom facilities, especially those providing a family room and/or changing area.
  - Service animal relief area.
  - Visual paging displays.

Mitigation Measures

- Provide training for all those interacting with the passenger with disabilities so they understand
  - How to best meet the needs of the passenger (e.g., wheelchair versus allowing the passenger to walk on their own with directions) and
  - The location of all vertical transitions, service animal relief areas, and accessible restroom facilities.
- Ensure that wayfinding information is clearly identified within the terminal, on the airport website, and in the airport app. Points of interest should include, at a minimum
  - Escalator and elevator access;
  - Restroom facilities, especially those providing a family room and/or changing area;
  - Service animal relief area; and
  - Visual paging displays.

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Security

Considerations

- Older adults and infrequent travelers may need assistance understanding what they must do to comply with security requirements.
- Upon check-in with the airline, passengers may be escorted through security to the airside by a third-party provider, airport staff member, or volunteer.

Mitigation Measures

- Provide clear directions to lanes for entry for the different needs of the passengers.
- Collaborate with the TSA to ensure that their staff members have been trained and aware of travelers with disabilities' different needs and capabilities.
- Work with the ADA coordinator to test the signage and directions with a passenger who is willing to test the system as a passenger when traveling through the airport.
- Offer an opportunity for the TSA staff to be trained along with the airport staff and business partners on providing services for travelers with disabilities and older adults.
- Provide volunteers or staff to assist travelers who may need extra help getting through security.



Airside

Considerations

- Once travelers are through security and in the airside area, wayfinding and directories are required to indicate the location of gates, food and beverage options, and retail locations. The information should meet ADA requirements for the person with vision impairment, hearing loss, and mobility issues. Standards for accessible design can be found at www.access-board.gov.
- Some passengers with disabilities may need assistance getting food before boarding their flight, finding an adult changing room, or locating a service animal relief area.
- Passengers with mobility issues may need assistance being wheeled or escorted from the gate area onto the plane.
- Passengers who are deaf or hard of hearing cannot hear announcements; therefore, the passenger may identify themselves so that the airline attendant can advise them of gate changes or delays either through mobile transmission or in person.
- Passengers who are blind or have low vision may need to identify themselves to the gate attendant so they can go to the passenger or call them to the podium to advise of any gate changes or delays.

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Mitigation Measures

- Wayfinding, directories, and flight information display systems need to be created with persons with disabilities in mind to ensure that the information is readable for all passengers. Flight Information Display systems must be at the appropriate angle with a large enough font so that passengers in wheelchairs can read the screens.
- Airport staff or third-party providers should not leave the passenger in a wheelchair with no contact information. They must advise the airline's gate attendant of the passenger's disabilities so they can assist the passenger as needed. Airport staff or third-party providers have a duty to assist the passenger with food and beverage needs, retail access, and restroom access.
- Wayfinding information is clearly identified within the terminal, on the airport website, and in the airport app. Points of interest should include, at a minimum
    - Escalator and elevator access,
    - Restroom facilities, especially those providing a family room and/or changing area,
    - Service animal relief area, and
    - Visual paging displays.



Return to Airport via Aircraft or Through Airside

Considerations

- Some displays indicating the connecting gate have a small font or change too fast for either older adults or passengers with cognitive disabilities to read.
- Often passengers requiring wheelchair assistance are left on the plane waiting for someone to assist. That may result in a long delay, which can be very stressful if the passenger has a close connection or needs to use the restroom.
- Signage in the immediate gate area should indicate the location of baggage claim and should be appropriate for those with a vision impairment. The airport map on the website or airport app should include this information as well.
- Passengers with vision impairment may not be able to recognize their baggage. In addition, they may need help lifting their bag off the baggage conveyor.

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Mitigation Measures

- Assign staff or volunteers to gate areas for arriving flights to assist passengers in identifying connecting gates or give directions to baggage claim.
- Airports should collaborate further with the airline and third-party providers to ensure passengers who need wheelchairs are provided with them as soon as they arrive at the terminal. Airport staff and volunteers may be able to assist with wheelchairs (depending upon the arrangement with the airport, airline, and third-party providers) or guide passengers to the baggage claim.
- Ensure that wayfinding information is clearly identified within the terminal, on the airport website, and in the airport app, showing the following:
  - Escalator and elevator access.
  - Restroom facilities, especially those providing a family room and/or changing area.
  - Service animal relief area.
  - Visual paging displays.
  - Baggage claim.
- Assign staff or volunteers to the baggage claim area who can assist passengers to identify or lift bags off the baggage carousel.



Departure from Airport

Considerations

- Signage for all forms of ground transportation must be clear for passengers who are being picked up. Points of interest should include the location of car rentals, bus stops, train location, shuttle locations, and TNC pickup points.
- Third-party wheelchair providers must coordinate with the person being served to understand how they are leaving the airport.

Mitigation Measures

- Signage should be provided throughout the passenger's journey immediately after exiting baggage claim and at various points throughout the journey. Review the signage with the ADA coordinator and/or a representative of the community of people with disabilities to ensure that the wayfinding meets the traveler's needs.
- Points of interest should be clearly identified through the airport website and mobile app.
- All staff, volunteers, and third-party providers receive training to understand how best to serve the passenger and their needs as they depart the airport.

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APPENDIX B

Communicate with the Passenger

The following section provides additional detail on the best practices for communicating with the passenger. Consider this general guidance related to the use of communication methods as you consider the implementation of specific touchless solutions.

Thinking about the airport journey from a wayfinding perspective, the journey really begins before the traveler leaves home. The remote processing of boarding passes and the use of digital wayfinding tools to anticipate the journey all contribute to the wayfinding experience. Once a traveler approaches the airport, this moving from the preplanned journey transforms to the physical environment. Due to the dynamic nature of environments that are changing due to modified passenger processing, travelers rely on communication hubs that support each decision point. Below are some key elements of the journey.

Parking and Arrival

Approach Roads

Travelers approaching the airport make many decisions. Whether parking, returning a rental car, dropping off or picking up, all require communication to the distinct routing. Considerations for dynamic operational requirements supported by a flexible, digital communication system may become part of this process. Overhead signs using full-matrix boards controlled by systems facilitate items such as queue management and flexible airline counter assignment.



Minneapolis-St. Paul International Airport Roadways.

Parking

Decisions around parking can be considered on the roadway system. Long-term, short-term, premium, and economy parking products all require users to make decisions before entering the garage. Within the garage, drivers want to find a parking spot as quickly as possible, then be reminded of where they parked,

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followed by the most efficient directions to the terminal. The use of a "smart" garage wayfinding product can accomplish the initial task with great efficiency.



Parking Signage (photo by TKH Security/Park Assist).



Jewel Changi Singapore Signage.

Curbside and Terminal Entrances

Like the approach roads, curbside dynamic signs can extend the flexible communication method needed to allow for varying operations of the airport. This wayfinding element can effectively direct travelers to the most appropriate entrance that suits the operational condition.



Minneapolis-St. Paul International Airport Curb.

Landside

Check-In

At the airport check-in hall, being able to communicate information on queuing, baggage drop-off, and other specific processes can be enabled in a wide variety of methods. A "ribbon" of communication can provide linear orientation to the passengers. This, in collaboration with the dynamic roadway and curbside systems, can create the greatest flexibility for communication. The integration of flight information, check-in information, and sense of place can be accommodated within check-in structures in a compelling manner.

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Harry Reid International Airport Signage *(photo by Clark County Department of Aviation).*



Louis Armstrong New Orleans International Airport Signage *[photo by Jonathan Cutrer on Flickr, (CC BY-NC 2.0)].*



Security

Identifying security points and informing travelers of wait times can help with queue management and overall traveler expectations. These are also opportunities to communicate other operational messaging.



Orlando InternationalAirport West Checkpoint Video Wall *(photo by Synect).*



Harry Reid International Airport: Synect's ReadySeeGo at the Innovation Checkpoint *(photo by Synect).*



Airside

Gates

Gate signs can go beyond simply identifying the gate number. These elements can also provide specific destination information, weather, flight status, or sense of place imagery.

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Harry Reid International Airport Gate Signage
(photo by the Clark County Department of Aviation).

Flight Information

Flight information displays placed at key decision points and areas where passengers have dwell time can be another method for clear communication. The scale of these elements can depend on the architectural environment.



Calgary International Airport Flight Information Displays.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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